

Dual Integral Quadratic Constraints for Robust Control of Partial Integral Equations

Thijs Lenssen¹

Control Systems Group, Eindhoven University of Technology, The Netherlands

Abstract

Partial Differential Equations (PDEs) model spatially distributed systems whose control must account for uncertainty and nonlinearities. Partial Integral Equations (PIEs) provide an operator representation supporting convex nominal H_∞ state-feedback synthesis. We extend this approach to robust control using Integral Quadratic Constraints (IQCs). Synthesis introduces bilinear couplings between the controller, storage, and multiplier operators. Combining induced L_2 -gain duality with hard J -spectral factorization, we establish a dual IQC representation. For an admissible factorization, this yields a convex Linear PI Inequality (LPI) that permits recovery of a controller, in closed form whenever the storage operator is coercive and separable, with guaranteed closed-loop robustness. For multiplier classes admitting a hard J -spectral factorization, including a spatially weighted sector class on infinite-dimensional channels, constructed here, the method accommodates infinite-dimensional uncertainty channels and dynamic filters.

Keywords: PDEs, Robust Control, LPIs, IQCs, PIEs

1. Introduction

Partial Differential Equations (PDEs) provide fundamental models of physical phenomena evolving in space and time. Seminal examples include Fourier's theory of heat conduction [1], d'Alembert's wave equation for vibrating strings [2], and Stokes' equations for viscous fluid motion [3]. These models form a basis for understanding thermal, mechanical, and fluid processes across science and engineering. Controlling these processes requires stability and performance guarantees that account for uncertain parameters, changing operating conditions, and nonlinear effects. Establishing these guarantees is challenging because both the dynamics and the uncertain or nonlinear interactions may be distributed over space.

A systematic treatment of these distributed dynamics begins with infinite-dimensional systems theory, which provides foundations for studying PDE solutions, input-output maps, and well-posed feedback interconnections [4–6]. Lasiecka and Triggiani [7] developed continuous and approximation theories for optimal control of abstract parabolic systems. Developments in robust stabilization and H_∞ control [8, 9] are complemented by backstepping and Lyapunov-based designs [10, 11], and semidefinite methods for PDE stability analysis [12]. A complementary computational route uses established robust-control

tools for Ordinary Differential Equations (ODEs), including frequency-domain methods, Riccati equations, and Linear Matrix Inequalities (LMIs) [13–17]. These tools apply to PDEs through spatial discretization, modal truncation, or lumped-parameter approximation. Although convergence and performance guarantees are available under suitable conditions [18], neglected modes can affect the closed loop through spillover [19]. This motivates computational methods that retain the distributed dynamics when constructing control certificates.

To retain distributed dynamics while exploiting such computational tools, the Partial Integral Equation (PIE) representation [20, 21] generalizes ODE state-space models to coupled finite-dimensional and distributed states. For supported PDE classes, it exactly incorporates the dynamics and boundary conditions into bounded Partial Integral (PI) operators. Their algebra is closed under addition, composition, and adjoints, enabling extensions of matrix-based ODE methods. Linear PI Inequalities (LPIs) generalize LMIs, and PIETOOLS [22, 23] converts parameterized LPI conditions into semidefinite programs while retaining the PIE plant representation. An industrial application is the PIE based observer design for the digital twin of an inkjet printer's thermo-fluidic fixation process developed and validated by Toolhally et al. [24]. Recently, Shivakumar et al. [25] established PIE duality and convex LPI conditions for nominal H_∞ -optimal state-feedback synthesis. This motivates extending PIE duality to uncertainty and nonlinearities described by structural information beyond unstructured small-gain conditions.

This approach builds on Lur'e and Postnikov's absolute-

Email address: t.m.lenssen@student.tue.nl, thijs.lenssen14@gmail.com (Thijs Lenssen)

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stability problem [26], Popov's frequency-domain criterion [27], the circle criterion, and Zames's input-output stability theory [28]. Zames–Falb multipliers [29] exploit monotonicity and slope restrictions. Megretski and Rantzer [30] established the modern Integral Quadratic Constraint (IQC) framework, unifying such descriptions for analysis of linear systems interconnected with uncertain or nonlinear components. Within the PIE framework, hard IQCs provide robust stability and performance tests for coupled ODE–PDE systems, including infinite-dimensional uncertainty channels [31–33]. Subsequent results connect PIE IQCs to structured singular-value bounds and robust observer with known uncertainty synthesis [34]. We develop a dual IQC representation for these analysis certificates and apply it to convex state-feedback synthesis.

We express these certificates in the language of Willems's dissipation theory [35]. For finite-dimensional linear systems, the famous Kalman–Yakubovich–Popov (KYP) lemma [36] connects frequency-domain inequalities to quadratic dissipation inequalities expressed as LMIs. Hard factorizations connect IQCs to dissipativity by expressing them as integral inequalities valid over every finite time horizon. This allows IQC analysis to be formulated through dissipation inequalities. Goh [37] linked IQC to generalized sectors, dissipativity and J -spectral factorization. For rational IQC multipliers, existence conditions for J -spectral factors are given by the canonical factorization theorem of Bart, Gohberg, and Kaashoek [38] and by Meinsma's characterization using equalizing vectors [39]. Seiler [40] used the latter to establish existence for strict positive–negative multipliers and showed that these factorizations yield hard IQCs and dissipation inequalities with nonnegative storage, related developments include doubly-hard factorizations and terminal-cost IQCs [41, 42]. For fixed filters and controllers, these certificates yield convex analysis conditions.

Designing the state-feedback gain introduces a synthesis obstacle already visible for ODEs. For a fixed IQC filter, consider the augmented dynamics

$$\dot{x} = Ax + Bw + B_u u, \quad v = Cx + Dw + D_{zu} u,$$

where v enters the IQC. Under state feedback $u = Kx$, a sufficient dissipation condition is

$$\begin{bmatrix} (P(A + B_u K)) + (P(A + B_u K))^T & PB \\ B^T P & \epsilon I \end{bmatrix} + \begin{bmatrix} (C + D_{zu} K)^T \\ D^T \end{bmatrix} V [C + D_{zu} K \quad D] \preceq 0,$$

where $P = P^T \succeq 0$ is the storage matrix, $V = V^T$ is the multiplier weight, and $\epsilon > 0$. For fixed K , this is an LMI. Optimizing K couples it to both P and V through $PB_u K$ and $VD_{zu} K$. The standard ODE invertible congruence transformation removes the first coupling but generally leaves the second. Existing methods use additional

structure or iterative design [43]. Dual IQCs also permit convex estimator and feedforward synthesis [44, 45], there by an elimination argument.

Our main result is a dual IQC construction through J -spectral factorization, combined with PIE duality. The construction supports infinite-dimensional channels and memoryless or dynamic filters. As an application, we derive a convex LPI condition for robust state-feedback synthesis with explicit controller recovery. The resulting controller guarantees stability and the prescribed quadratic performance for all uncertainties satisfying the IQC.

The paper is organized as follows. Sections 2,3 introduce the preliminaries, problem definition and standing assumptions. Section 4 establishes robust stability and performance through the IQC framework. Section 5 examines the primal analysis condition, and Sections 6–7 develop duality and convex controller synthesis. Sections 8 and 9 present numerical examples and connections to finite-dimensional dynamic multipliers. Section 10 concludes the paper, proofs are collected in the appendix.

2. Notation & Preliminaries

2.1. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^T y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$. We omit dimensions when clear from context. For suitably differentiable spatial states, $\dot{\mathbf{x}} := \partial_t \mathbf{x}$. We denote the induced norm for bounded linear operators on $\mathbb{R}L_2$ by

$$\|\mathcal{P}\| := \sup_{0 \neq \mathbf{x} \in \mathbb{R}L_2^n} \frac{\|\mathcal{P}\mathbf{x}\|_{\mathbb{R}L_2}}{\|\mathbf{x}\|_{\mathbb{R}L_2}}.$$

Our notational convention is to write spatial states and profiles in bold (e.g., $\mathbf{x} \in \mathbb{R}L_2$). For an operator $\mathcal{A}$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_{\mathbb{R}L_2} = \langle \mathcal{A}^* \mathbf{x}, \mathbf{y} \rangle_{\mathbb{R}L_2}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}L_2.$$

Define the signal space

$$\begin{aligned} L_{[a,b]}^{\mathbf{n}} &:= \{\mathbf{u} : [0, \infty) \rightarrow \mathbb{R}L_2^{\mathbf{n}}[a, b] \mid \\ &\quad \|\mathbf{u}\|_{L_{[a,b]}^2}^2 := \int_0^\infty \|\mathbf{u}(t)\|_{\mathbb{R}L_2}^2 dt < \infty\}, \end{aligned}$$

with inner product $\langle \mathbf{u}, \mathbf{v} \rangle_{L_{[a,b]}} := \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbb{R}L_2} dt$. For any $\mathbf{y} : [0, \infty) \rightarrow \mathbb{R}L_2^n[a, b]$, define the truncation operator $(p_T \mathbf{y})(t) := \mathbf{y}(t)$ for $0 \leq t \leq T$ and $(p_T \mathbf{y})(t) := 0$ for $t > T$, the corresponding extended signal space is therefore,

$$L_{e,[a,b]}^{\mathbf{n}} := \{\mathbf{v} \mid p_T \mathbf{v} \in L_{[a,b]}^{\mathbf{n}} \forall T > 0\}.$$

Moreover, for $\tau \geq 0$ define the shift operator $(s_\tau u)(t) := u(t - \tau)$. An operator $G : L_{e,[a,b]}^{\mathbf{n}} \rightarrow L_{e,[a,b]}^{\mathbf{m}}$ is called

- *Causal*, if $p_T G = p_T G p_T$ for all $T > 0$.
- *Anticausal*, if $(I - p_T)G(I - p_T) = (I - p_T)G$ for all $T > 0$.
- *Memoryless*, if G is simultaneously causal and anticausal.
- *Time-Invariant*, if $s_\tau G = G s_\tau$ for all $\tau \geq 0$.
- *Bounded on $L_{[a,b]}$* , if $\|G\| \leq C$ for some $C \geq 0$.
- *Bounded on $L_{e,[a,b]}$* , if $\|p_T G\| \leq C$ for some $C \geq 0$ and all $T > 0$.

For a bounded linear operator $\mathcal{K} : \mathbb{R}L_2^{\mathbf{n}} \rightarrow \mathbb{R}L_2^{\mathbf{m}}$, define the multiplication operator $\mathcal{M}_{\mathcal{K}} : L_{e,[a,b]}^{\mathbf{n}} \rightarrow L_{e,[a,b]}^{\mathbf{m}}$ by

$$(\mathcal{M}_{\mathcal{K}} u)(t) := \mathcal{K}u(t).$$

2.2. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are integral operators defined jointly on a vector space and Hilbert space [20, 21].

Definition 1. We say an operator $\mathcal{P} : \mathbb{R}L_2^{\mathbf{n}}[a, b] \rightarrow \mathbb{R}L_2^{\mathbf{m}}[a, b]$ belongs to $\Pi_4^{\mathbf{m} \times \mathbf{n}}$, denoted by $\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R_0, R_1, R_2\} \end{array} \right]$, if there exist a matrix P and bounded measurable matrix-valued functions Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\left[\left(\Pi \left[\begin{array}{c|c} P & Q_1 \\ \hline Q_2 & \{R\} \end{array} \right] \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathbf{R} \mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R} \mathbf{x})(s) = R_0(s) \mathbf{x}(s) + \int_a^s R_1(s, \theta) \mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta) \mathbf{x}(\theta) d\theta.$$

where $\mathcal{R} \in \Pi_3$.

The vector space of PI operators, given compatible dimensions, is closed under composition, addition, adjoint and concatenation – implying that Π_4 is a $*$ -algebra [21].

Definition 2. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\ell(\cdot)$, a

linear PI inequality is a convex optimization of the following form

$$\begin{aligned} & \min_{\substack{P_{ij}, Q_{1ij}, Q_{2ij}, \\ R_{0ij}, R_{1ij}, R_{2ij} \}_{j=1}^K} \ell(\{P_{ij}, Q_{1ij}, Q_{2ij}, R_{0ij}, R_{1ij}, R_{2ij}\}) \\ \text{s.t. } & \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \left[\begin{array}{c|c} P_{ij} & Q_{1ij} \\ \hline Q_{2ij} & \{R_{ij}\} \end{array} \right] \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0. \end{aligned}$$

An LPI is solved by reducing it to an LMI, for which standard semidefinite programming applies. We next describe PIE systems and their stability properties, following [25]. Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}[a, b]$ and $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T} \mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix}$$

for all $t \geq 0$ and $\mathcal{T} \mathbf{x}(0) = \mathcal{T} \mathbf{x}_0$. Additionally, for signals $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}[a, b]$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}[a, b]$ and $\underline{z}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}[a, b]$ satisfy the dual PIE, $G^\top$, with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^* \underline{\mathbf{x}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix},$$

for all $t \geq 0$ and $\mathcal{T}^* \underline{\mathbf{x}}(0) = \mathcal{T}^* \underline{\mathbf{x}}_0$. A notational convention is to underline anything related to the dual system. For these systems, we consider input–output and exponential stability. Given a PIE system G , we say G is input–output stable if for any $z, w \in L_{e,[a,b]}$ satisfying $z = Gw$ with $\mathbf{x}_0 = 0$, we have

$$\|p_T z\|_{L_{[a,b]}} \leq \gamma \|p_T w\|_{L_{[a,b]}}, \quad \forall T \geq 0, \quad \forall w \in L_{e,[a,b]}^{\mathbf{n}_w}.$$

Given a PIE system G , we say that the homogeneous part of G is *exponentially stable* with decay rate $\alpha > 0$ if there exists some $M > 0$ such that for any $\mathbf{x}_0 \in \mathbb{R}L_2$, if $\mathcal{T} \mathbf{x}(0) = \mathcal{T} \mathbf{x}_0$ and $\mathcal{T} \dot{\mathbf{x}}(t) = \mathcal{A} \mathbf{x}(t)$, then

$$\|\mathcal{T} \mathbf{x}(t)\|_{\mathbb{R}L_2} \leq M \|\mathbf{x}_0\|_{\mathbb{R}L_2} e^{-\alpha t} \quad \forall t \geq 0.$$

To describe feedback interconnections, we use the following notation. For compatible causal G and H , the Redheffer star products denote the well-posed feedback interconnection obtained by eliminating internal signals, such that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = G \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \begin{array}{ll} G \star H : u_2 \mapsto y_2 & u_1 = H y_1 \\ H \star G : u_1 \mapsto y_1 & u_2 = H y_2 \end{array}.$$

provided the respective interconnections are well-posed. We say that the interconnection $G \star H$ is *well-posed* if

- *Existence and Uniqueness*, For any $u_2 \in L_{e,[a,b]}$, there exist unique $y_2 \in L_{e,[a,b]}$, $y_1 \in L_{e,[a,b]}$ and $u_1 \in L_{e,[a,b]}$ such that y_2, u_2, y_1, u_1 satisfy the interconnection.

- *Causality*, If y_1, u_1 and $\hat{y}_1, \hat{u}_1$ satisfy the interconnection for inputs u_2 and $\hat{u}_2$ with $p_T u_2 = p_T \hat{u}_2$, then $p_T(y_1 - \hat{y}_1) = 0$ and $p_T(u_1 - \hat{u}_1) = 0$, for all $T \geq 0$.

For dual PIE systems we can define the following operations

$$(G_1 G_2)^\top = G_2^\top G_1^\top, \quad (G_1 + G_2)^\top = G_1^\top + G_2^\top, \\ (G^\top)_{ij} = (G_{ji})^\top, \quad (G \star H)^\top = G^\top \star H^\top,$$

and $(G^{-1})^\top = (G^\top)^{-1}$ whenever the inverse exists.

2.3. PIETOOLS

PIETOOLS is a MATLAB toolbox developed by Shivakumar, Das, and Peet [22] for declaring, manipulating, and solving LPIs. Its operator syntax, PDE-to-PIE conversion, and optimization workflows are documented by Shivakumar et al. in the user manual [23]. See also the PIETOOLS homepage.

3. Problem Definition

A recent state-space framework based on Partial Integral Equations (PIEs) provides a formulation of stabilization and H_∞ -optimal state-feedback controller synthesis for PDEs [25]. This framework addresses the nominal setting, corresponding to $\Delta = 0$. Building on this result, we develop a framework for optimal state-feedback control of uncertain or nonlinear PIEs subject to a prescribed quadratic performance requirement. Let G denote the open-loop PIE plant to which the controller is connected. For a state-feedback controller K , the closed-loop interconnection $K \star G \star \Delta$ has the realization

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} + \mathcal{B}_u K & \mathcal{B} \\ \mathcal{C} + \mathcal{D}_{zu} K & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \\ w_\Delta = \Delta z_\Delta$$

where $z(t) = \begin{bmatrix} z_\Delta(t) \\ z_p(t) \end{bmatrix}$, and $w(t) = \begin{bmatrix} w_\Delta(t) \\ w_p(t) \end{bmatrix}$. Our objective is to construct a state-feedback controller K such that $K \star G \star \Delta$ is robustly stable and satisfies the prescribed performance requirement for every $\Delta \in \mathbf{\Delta}$. Throughout the paper, we make the following *standing assumptions*

- (A.1) $K \star G \star \Delta$ is well-posed for every $\Delta \in \mathbf{\Delta}$.
- (A.2) Robust stability and performance are analyzed with $\mathbf{x}(0) = 0$.
- (A.3) The controller has access to the PIE state $\mathbf{x}$. For a dynamic-filter controller, $z_\Delta, w_\Delta = \Delta z_\Delta$, and w_p are also available to the filter Θ , which is what a dynamic factor acts on.
- (A.4) Exogenous inputs and uncertainty do not act directly on the boundary [25, Sec. III-B]. Boundary channels satisfy $\dot{w} = \hat{w}$, where w is available for state reconstruction. For boundary control, $\dot{u}$ is generated by the controller and $u(t) = u(0) + \int_0^t \dot{u}(\tau) d\tau$ is the actual input to the plant.

The next section provides us with the tool we use to assess robust stability and performance.

4. Infinite Dimensional IQC Theory

This section introduces the IQC objects used to study stability and performance of interconnected systems in a Hilbert space. The construction builds on the classical IQC framework [30] and its extension to hard IQCs for infinite-dimensional channels [32]. We establish the generalized system interconnection, formalize stability and performance, and define the uncertainty and performance IQCs, as described in [34]. These definitions provide the foundation for the theory developed in sections 5, 6, and 7.

4.1. Description of uncertain systems in a Hilbert-space

We consider operators that map external inputs to external outputs through a feedback interconnection between linear (infinite dimensional) operators P and Δ . This interconnection structure is formally defined as follows

Definition 3 (Generalized Plant). *Suppose $P_\Delta : L_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$, $P_{\Delta p} : L_{e,[a,b]}^{\mathbf{n}_{w_p}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$, $P_{p\Delta} : L_{e,[a,b]}^{\mathbf{n}_{w_\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_p}}$, and $P_p : L_{e,[a,b]}^{\mathbf{n}_{w_p}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_p}}$, are bounded operators and $\Delta : L_{e,[a,b]}^{\mathbf{n}_{z_\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$, define*

$$\begin{bmatrix} z_\Delta \\ z_p \end{bmatrix} = P \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta p} \\ P_{p\Delta} & P_p \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (1)$$

for input $w_p \in L_{e,[a,b]}^{\mathbf{n}_{w_p}}$ and output $z_p \in L_{e,[a,b]}^{\mathbf{n}_{z_p}}$, there exist internal signals $z_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{z_\Delta}}$ and $w_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{w_\Delta}}$ satisfying the interconnected system $z_p = (P \star \Delta)w_p$.

The interconnection structure described above captures the feedback interaction between the nominal system P and the uncertainty Δ , where w_p and z_p represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs. This is the requirement recorded in (A.1). Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 3.

4.2. Robust stability and performance

Veenman, Scherer, and K ro glu [46] describe both uncertainty and performance specifications through quadratic constraints on interconnection signals. For PIE systems, this formulation also supports structured robustness and performance analysis [32, 34]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of

the performance channels in (1), and introducing auxiliary signals z_0 and w_0 .

Definition 4. *The interconnection*

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0, \quad (2)$$

with $z_0, w_0 \in L_{e,[a,b]}$, is said to be robustly stable if

1. The uncertain system is well-posed for all $\Delta \in \mathbf{\Delta}$.
2. The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $L_{e,[a,b]}$ for all $\Delta \in \mathbf{\Delta}$.

Within the IQC framework, the uncertainty operator Δ is described by a quadratic constraint defined by an appropriate multiplier. A stability certificate combines this constraint with a complementary condition on the plant. We use the finite-horizon, or hard, IQC formulation [32, 40], leading to the following definition.

Definition 5. An uncertainty operator $\Delta : L_{e,[a,b]}^{\mathbf{n}_{z\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{w\Delta}}$ is said to satisfy the hard IQC $(\Psi_\Delta, \mathcal{V}_\Delta)$ defined by the causal, bounded dynamic operator $\Psi_\Delta : L_{e,[a,b]}^{\mathbf{n}_{z\Delta} + \mathbf{n}_{w\Delta}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z\Delta} + \mathbf{n}_{w\Delta}}$ and linear bounded $\mathcal{V}_\Delta = \mathcal{V}_\Delta^*$ if the inequality holds for all $z_\Delta \in L_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ \Delta z_\Delta \end{bmatrix} \right\rangle_{L_{[a,b]}} \geq 0.$$

Moreover, let $\mathcal{V}_\Delta\{\mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta\}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied. The associated input-output maps are evaluated from the zero initial condition in (A.2).

Definition 6. Robust performance is achieved if the interconnection $P \star \Delta$ is robustly stable and there exists $\epsilon > 0$ such that, for performance IQC $(\Psi_p, \mathcal{V}_p)$ defined by the causal, bounded operator $\Psi_p : L_{e,[a,b]}^{\mathbf{n}_{z_p} + \mathbf{n}_{w_p}} \rightarrow L_{e,[a,b]}^{\mathbf{n}_{z_p} + \mathbf{n}_{w_p}}$ and $\mathcal{V}_p\{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\} = \mathcal{V}_p^*$, such that

$$\left\langle p_T \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} v \\ 0 \end{bmatrix} \right\rangle_{L_{[a,b]}} \geq 0$$

for any $v \in L_{e,[a,b]}^{\mathbf{n}_{z_p}}$ we have

$$\begin{aligned} & \left\langle p_T \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P \star \Delta \\ I \end{bmatrix} w_p \right\rangle_{L_{[a,b]}} \\ & \leq -\epsilon \|p_T w_p\|_{L_{[a,b]}}^2. \end{aligned}$$

for all $w_p \in L_{e,[a,b]}^{\mathbf{n}_{w_p}}$ and $T \geq 0$.

In this framework, the multipliers $\mathcal{M}_{\mathcal{V}_\Delta}$ and $\mathcal{M}_{\mathcal{V}_p}$ serve distinct but complementary roles. The hard uncertainty IQC $(\Psi_\Delta, \mathcal{M}_{\mathcal{V}_\Delta})$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the performance IQC $(\Psi_p, \mathcal{M}_{\mathcal{V}_p})$ encodes performance requirements for the nominal system, such as L_2 -gain, passivity conditions, or frequency weighted input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives. This section applies the IQC framework to the primal system and gives a certificate for simultaneous robust stability and performance. For brevity, we write the hard IQC as

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2.$$

Let Ψ_Δ and Ψ_p have a 2×2 block partition and define $\Psi_{ij} := \text{diag}((\Psi_\Delta)_{ij}, (\Psi_p)_{ij})$ for $i, j \in \{1, 2\}$. Let $w^\top = [w_\Delta^\top \ w_p^\top]$ and $z^\top = [z_\Delta^\top \ z_p^\top]$. Moreover, define $\mathcal{Q} = \text{diag}(\mathcal{Q}_\Delta, \mathcal{Q}_p)$, $\mathcal{S} = \text{diag}(\mathcal{S}_\Delta, \mathcal{S}_p)$, and $\mathcal{R} = \text{diag}(\mathcal{R}_\Delta, \mathcal{R}_p)$, then $\mathcal{V}\{\mathcal{Q}, \mathcal{S}, \mathcal{R}\}$.

Theorem 1 (Robust Stability and Performance). *Let $P, \mathbf{\Delta}$ be given, and there exists $\Psi_\Delta, \mathcal{V}_\Delta$, as in Definition 5, such that,*

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \mathbf{\Delta}$.
- Any uncertainty $\Delta \in \mathbf{\Delta}$ satisfy the hard IQC defined by Ψ_Δ and $\mathcal{V}_\Delta$.

Then the interconnection $P \star \Delta$ is robustly stable and robust performance is achieved with respect to some Ψ_p and $\mathcal{V}_p = \{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\}$ as in Definition 6, if there exists $\epsilon > 0$ such that,

$$\left\langle p_T \Psi \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a,b]}} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2 \quad (3)$$

for all $w_\Delta \in L_{e,[a,b]}^{\mathbf{n}_{w\Delta}}$, $w_p \in L_{e,[a,b]}^{\mathbf{n}_{w_p}}$ and $T \geq 0$.

Proof. See Appendix A.1. ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system. We next specialize this certificate to the controlled primal realization and expose bilinearities in the LPI that distinguish convex analysis from convex controller synthesis.

5. Controller Analysis (Primal System)

The IQC certificate of Section 4 becomes a controller analysis condition once the state-feedback law is fixed. We examine the primal control scheme shown in Figure 2,

$$\begin{aligned}
G &:= \left\{ \begin{bmatrix} \mathcal{T}_G \dot{\mathbf{x}}_G(t) \\ z(t) \\ y_G(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G & \mathcal{B}_G & \mathcal{B}_u \\ \mathcal{C}_G & \mathcal{D}_G & \mathcal{D}_{zu} \\ I & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ w(t) \\ u(t) \end{bmatrix} \right. \\
K &:= \left\{ u(t) = [\mathcal{K}_G \quad \mathcal{K}_\Theta] \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \end{bmatrix} \right. \\
\Theta &:= \left\{ \begin{bmatrix} \mathcal{T}_{\dot{\mathbf{x}}_\Theta}(t) \\ \tilde{v}(t) \\ y_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \\ I & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ v(t) \end{bmatrix} \right. \quad \text{with } v(t) = \begin{bmatrix} z(t) \\ w(t) \end{bmatrix}, \\
\Theta \left[\begin{smallmatrix} K \star G \\ I \end{smallmatrix} \right] &:= \left\{ \begin{bmatrix} \mathcal{T}_G \dot{\mathbf{x}}_G(t) \\ \mathcal{T}_{\dot{\mathbf{x}}_\Theta}(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G + \mathcal{B}_u \mathcal{K}_G & \mathcal{B}_u \mathcal{K}_\Theta & \mathcal{B}_G \\ \mathcal{B}_1(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{A} + \mathcal{B}_1 \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{B}_1 \mathcal{D}_G + \mathcal{B}_2 \\ \mathcal{D}_{11}(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{C}_1 + \mathcal{D}_{11} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{11} \mathcal{D}_G + \mathcal{D}_{12} \\ \mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G) & \mathcal{C}_2 + \mathcal{D}_{21} \mathcal{D}_{zu} \mathcal{K}_\Theta & \mathcal{D}_{21} \mathcal{D}_G + \mathcal{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix} \right\} \quad (4)
\end{aligned}$$

Figure 1: Primal components, G denotes the nominal part of the plant, K denotes the state feedback controller, Θ denotes the J-spectral factor of the IQC. $\Theta \begin{bmatrix} K \star P \\ I \end{bmatrix}$ denotes the primal interconnection.

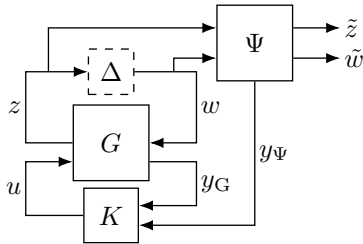


Figure 2: Primal controller interconnection.

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \tilde{v}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} & \mathcal{B}_u \\ \mathcal{C} & \mathcal{D} & \mathcal{D}_{zu} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \\ u(t) \end{bmatrix}. \quad (5)$$

A sufficient condition for robust stability and performance, [34], is the existence of a PI operator $\mathcal{P} = \mathcal{P}^* \succeq 0$ and $\epsilon > 0$ satisfying

$$\begin{aligned} & \begin{bmatrix} \mathcal{T}^* \mathcal{P}(\mathcal{A} + \mathcal{B}_{\text{u}} \mathcal{K}) + (\mathcal{A} + \mathcal{B}_{\text{u}} \mathcal{K})^* \mathcal{P} \mathcal{T} & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} \\ & + \begin{bmatrix} (\mathcal{C} + \mathcal{D}_{\text{zu}} \mathcal{K})^* \\ \mathcal{D}^* \end{bmatrix} \mathcal{V} \begin{bmatrix} \mathcal{C} + \mathcal{D}_{\text{zu}} \mathcal{K} & \mathcal{D} \end{bmatrix} \preceq 0. \end{aligned} \quad (6)$$

We consider a generalized plant P with a PIE state-space realization. The nominal PIE duality results of Shiva Kumar et al. [25] provide the starting point for convex state-feedback synthesis. For dynamically filtered IQCs, two further ingredients are relevant, hard factorizations connecting IQC theory to dissipativity theory [40, 42, 47], and the dual system representations developed for LPV estimation and feedforward synthesis [44, 45]. This section develops the corresponding dual-filter construction for PIE systems using the induced L_2 -gain PIE duality result.

6.1. Nominal PIE induced L_2 -norm duality

We begin by restating the main theorem from [25]. In [25], the theorem is formulated for the case where $\mathcal{D}$ is a matrix and the induced L_2 norm is bounded over an infinite time horizon. Generalizing this result to the case where $\mathcal{D}$ is a PI operator and considering the bounded truncated induced $L_{[a,b]}$ norm does not alter the argument.

Theorem 2 (Dual L_2 -gain [25, Thm. 9]). *For a PIE system with corresponding PI operators $P\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$, the following statements are equivalent.*

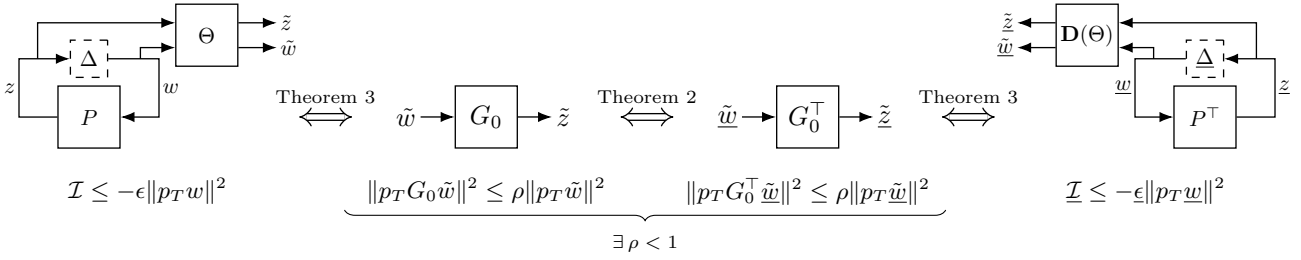


Figure 3: Equivalence chain for primal and dual dynamic filtering.

1. For any $w \in L_{e,[a,b]}^{\mathbf{n}_w}$, and $\mathbf{x}(0) = 0$, P satisfies $\|p_T z\| \leq \gamma \|p_T w\|$ for all $T > 0$.
2. For any $\underline{w} \in L_{e,[a,b]}^{\mathbf{n}_z}$, and $\underline{\mathbf{x}}(0) = 0$, $P^\top$ satisfies $\|p_T \underline{z}\| \leq \gamma \|p_T \underline{w}\|$ for all $T > 0$.

For a bounded, time-invariant, memoryless linear uncertainty Δ , suppose the feedback interconnection $P \star \Delta$ is well-posed, in accordance with (A.1). Closing the loop $w = \Delta z$ gives

$$\mathcal{T}\dot{\mathbf{x}} = (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})\mathbf{x}.$$

Taking adjoints and using the push-through identity yields

$$\begin{aligned} (\mathcal{T})^* &= \mathcal{T}^*, \\ &= (\mathcal{A} + \mathcal{B}\Delta(I - \mathcal{D}\Delta)^{-1}\mathcal{C})^* \\ &= \mathcal{A}^* + \mathcal{C}^*\Delta^*(I - \mathcal{D}^*\Delta^*)^{-1}\mathcal{B}^*. \end{aligned}$$

This is exactly the state operator of $P^\top \star \Delta^*$. Thus, [25, Thm. 6] establishes equivalence of weak convergence for the primal and dual states, where weak convergence is convergence of $\mathcal{T}\mathbf{x}(t)$ to zero in the weak topology of $\mathbb{R}L_2$, that is $\langle \mathcal{T}\mathbf{x}(t), \mathbf{y} \rangle_{\mathbb{R}L_2} \rightarrow 0$ for every $\mathbf{y} \in \mathbb{R}L_2$. Exponential stability is likewise equivalent by [25, Cor. 8]. For the constant repeated real uncertainty $\Delta = \delta I$ in Section 8.2, $\Delta^* = \Delta$, so the same uncertainty structure appears in both interconnections, provided the performance IQC $(\Psi_p, \mathcal{V}_p)$ admits a J-spectral factorization.

6.2. Duality under dynamic filtering

For a general uncertainty Δ , this construction is less trivial. It is therefore more convenient to develop duality through the IQC description, whose filters and multiplier weights are linear operators, rather than through Δ itself. The inertia argument of [44] assumes linear uncertainty. Venkataraman and Seiler [45] accommodate nonlinear uncertainties through dual IQCs and J-spectral factorizations of strict Positive-Negative (PN) IQCs. We adopt the same strategy. We first reformulate the IQC condition as a strict small-gain requirement on an associated system G_0 with respect to the truncated $L_{[a,b]}$ norm, the extended space itself carrying no norm, then apply Thm.2, and finally reconstruct the dual system. In their framework, this relationship is established through congruence transformations of the analysis LMIs [45, Sec. 2-3]. Our approach

relies on the induced truncated $L_{[a,b]}$ -gain duality result. Let us define a hard J-spectral factorization.

Definition 7 (*J-Spectral Factorization and Dual Filter*). For a given IQC defined by $(\Psi, \mathcal{V})$ and a set Δ' of causal bounded operators, we say that a causal linear bounded operator Θ and $J_{\mathbf{n}_z, \mathbf{n}_w} := \begin{bmatrix} I_{\mathbf{n}_z} & 0 \\ 0 & -I_{\mathbf{n}_w} \end{bmatrix}$ (the pair $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$) is a hard J-spectral factorization of $(\Psi, \mathcal{V})$ relative to Δ' if,

- $\Theta^* \mathcal{M}_{J_{\mathbf{n}_z, \mathbf{n}_w}} \Theta = \Psi^* \mathcal{M}_{\mathcal{V}} \Psi$,
- Θ and Θ_{22} are causally boundedly invertible,
- the factorization is hard relative to Δ' : for every $\Delta' \in \Delta'$, every $z \in L_{e,[a,b]}$ and every $T \geq 0$,

$$\left\langle p_T \Theta \begin{bmatrix} z \\ \Delta' z \end{bmatrix}, p_T \mathcal{M}_{J_{\mathbf{n}_z, \mathbf{n}_w}} \Theta \begin{bmatrix} z \\ \Delta' z \end{bmatrix} \right\rangle_{L_{[a,b]}} \geq 0.$$

Following the dual-filter construction in [45], we use $\mathbf{D}(\Theta)$ to denote the dual filter associated with Θ of the form

$$\mathbf{D}(\Theta) := \left(\hat{J}^* \Theta^\top \hat{J} \right)^{-1},$$

where $\hat{J} = \hat{J}_{\mathbf{n}_z, \mathbf{n}_w} := \begin{bmatrix} 0_{\mathbf{n}_z \times \mathbf{n}_w} & I_{\mathbf{n}_z} \\ -I_{\mathbf{n}_w} & 0_{\mathbf{n}_w \times \mathbf{n}_z} \end{bmatrix}$. Write $\hat{J}_1 := \hat{J}_{\mathbf{n}_z, \mathbf{n}_w}$ and $\hat{J}_2 := \hat{J}_{\mathbf{n}_w, \mathbf{n}_z}$. The dual filter operation has the following properties,

1. *Multiplicativity*, $\mathbf{D}(\Theta_1 \Theta_2) = \mathbf{D}(\Theta_1) \mathbf{D}(\Theta_2)$ for invertible Θ_1, Θ_2 of compatible dimension. Indeed $\hat{J} \hat{J}^* = I$, so $\hat{J}^* (\Theta_1 \Theta_2)^\top \hat{J} = \hat{J}^* \Theta_2^\top \hat{J} \hat{J}^* \Theta_1^\top \hat{J}$, and inverting reverses the order again.
2. *Involutivity*, $\mathbf{D}(\mathbf{D}(\Theta)) = \Theta$, from $\hat{J}_1 \hat{J}_2 = -I$ and $(\Theta^\top)^\top = \Theta$.
3. *Scaling*, $\mathbf{D}(c\Theta) = c^{-1} \mathbf{D}(\Theta)$ for a nonzero scalar c , since $\mathbf{D}$ inverts its argument.

Remark 1 (Hard factorization requirement). For finite-dimensional channels, the hard-factorization requirement covers a broad class of multipliers [40, 42, 47]. In particular, strict PN multipliers, even when introduced through

soft frequency-domain IQCs, admit a hard J -spectral factorization [40, Thm. 4]. Strict PN is also preserved by the primal–dual transformation [45]. For infinite-dimensional channels, however, hardness of a proposed J -spectral factorization must be established explicitly. Callegari et al. [33] propose IQC lifting, which could extend finite-dimensional rational J -spectral factorizations to multipliers acting on infinite-dimensional channels.

For further brevity, define the primal quadratic form

$$\mathcal{I} := \left\langle p_T \Theta \begin{bmatrix} Pw \\ w \end{bmatrix}, p_T \mathcal{M}_{J_{n_z, n_w}} \Theta \begin{bmatrix} Pw \\ w \end{bmatrix} \right\rangle_{L_{[a, b]}}.$$

The corresponding dual form is evaluated on the graph of $P^\top$ using the dual filter $\mathbf{D}(\Theta)$ and the J_{n_w, n_z} operator,

$$\underline{\mathcal{I}} := \left\langle p_T \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix}, p_T \mathcal{M}_{J_{n_w, n_z}} \mathbf{D}(\Theta) \begin{bmatrix} P^\top \underline{w} \\ \underline{w} \end{bmatrix} \right\rangle_{L_{[a, b]}}.$$

The following theorem is our *main result* and establishes the system relations in Figure 3

Theorem 3 (Primal–dual IQC equivalence). *Let P be as in Definition 3, and let (Θ, J_{n_z, n_w}) be a hard J -spectral factorization. Assume that both P and Θ admit PIE realizations. Finally, assume $(\Theta_{21}P + \Theta_{22})$ has a bounded causal inverse. Define*

$$G_0 = (\Theta_{11}P + \Theta_{12})(\Theta_{21}P + \Theta_{22})^{-1}$$

then the following statements are equivalent,

1. *There exists some $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L_{[a, b]}} \leq \rho \|p_T \tilde{w}\|_{L_{[a, b]}}$ for every $\tilde{w}$ and $T \geq 0$.*
2. *There exists some $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L_{[a, b]}}^2$ for every w and $T \geq 0$.*
3. *There exists some $\underline{\epsilon} > 0$ such that $\underline{\mathcal{I}} \leq -\underline{\epsilon} \|p_T \underline{w}\|_{L_{[a, b]}}^2$ for every $\underline{w}$ and $T \geq 0$.*

Proof. See Appendix A.2. ■

The proof of Theorem 3 is operator-theoretic. It uses boundedness, causality, adjoints and the algebra of Section 2.

6.3. Dual robust-performance certificate

Theorem 3 establishes equivalence of the strict primal and dual IQC inequalities through a small-gain requirement on G_0 . Either side of the filtered graph may therefore be used to construct the same performance certificate. The resulting KYP condition permits the robust-performance certificate to be imposed on either the primal or dual realization.

Corollary 1. *Let P be as in Definition 3, and let (Θ, J_{n_z, n_w}) be a hard J -spectral factorization. Assume that both P and Θ admit bounded PIE realizations. Let $\Delta \in \Delta$ such that,*

- *The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \Delta$.*
- *The IQC in Definition 5 is satisfied for all $\Delta \in \Delta$.*
- *Let $\mathcal{D}, \mathcal{D}_P$ denote the feedthrough of Θ and P respectively, such that $(\mathcal{D}_{21}\mathcal{D}_P + \mathcal{D}_{22})^{-1} \in \Pi_4$.*

Suppose that either of the two statements hold for some $P = P^* \succeq 0 \in \Pi_4$ and a positive margin ϵ or $\underline{\epsilon}$, respectively.

- $$\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B} \\ \mathcal{B}^* \mathcal{P} \mathcal{T} & \epsilon I \end{bmatrix} + \begin{bmatrix} \mathcal{C}^* \\ \mathcal{D}^* \end{bmatrix} J_{n_z, n_w} \begin{bmatrix} \mathcal{C} & \mathcal{D} \end{bmatrix} \preceq 0$$
- $$\begin{bmatrix} \underline{\mathcal{T}}^* \underline{\mathcal{P}} \underline{\mathcal{A}} + (*) & \underline{\mathcal{T}}^* \underline{\mathcal{P}} \underline{\mathcal{B}} \\ \underline{\mathcal{B}}^* \underline{\mathcal{P}} \underline{\mathcal{T}} & \underline{\epsilon} I \end{bmatrix} + \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} J_{n_w, n_z} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0$$
 (8)

where, for any input $w \in L_{e, [a, b]}$ or $\underline{w} \in L_{e, [a, b]}$, the corresponding output z or $\underline{z}$ satisfies the primal or dual realization,

$$\Theta \begin{bmatrix} P \\ I \end{bmatrix} w := \left\{ \begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \right.$$

or

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} := \left\{ \begin{bmatrix} \underline{\mathcal{T}} \dot{\underline{\mathbf{x}}}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \right. \quad (9)$$

Then the interconnection $P \star \Delta$ with $\mathbf{x}(0) = 0$, as in (A.2), is robustly performing.

Proof. See Appendix A.3 ■

The next section applies duality to recover convex state-feedback synthesis.

7. Convex Controller Synthesis (Dual System)

Theorem 1 provides a robust-performance certificate for PIE systems in terms of the dual IQC. For analysis, this merely reformulates the primal certificate, since primal–dual equivalence preserves feasibility. The benefit of the dual formulation therefore appears in controller synthesis. Both controller constructions below use the same synthesis LPI, obtained by applying (8) to the corresponding closed-loop PIE. The choice of dual uncertainty filter changes only the realization of the controller recovered from this LPI. Related applications of IQC duality include the robust-estimation and feedforward result of Scherer and Köse [44, 48], and convex LPV synthesis of estimators and feedforwards [45].

$$\begin{aligned}
G^\top &:= \begin{cases} \begin{bmatrix} \mathcal{T}_G^* \dot{\mathbf{x}}_G(t) \\ \underline{z}(t) \\ \underline{y}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G^* & \mathcal{C}_G^* & I \\ \mathcal{B}_G^* & \mathcal{D}_G^* & 0 \\ \mathcal{B}_u^* & \mathcal{D}_{zu}^* & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \underline{w}(t) \\ \underline{u}_G(t) \end{bmatrix} & K^\top := \begin{cases} \begin{bmatrix} \underline{u}_G(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{K}_G^* \\ \mathcal{K}_\Theta^* \end{bmatrix} \underline{y}(t) \end{cases} \\
\mathbf{D}(\Theta) &:= \begin{cases} \begin{bmatrix} \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{v}}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}} & \underline{\mathcal{B}} & I \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* - \mathcal{C}^* \hat{J} \mathbf{D}(\mathcal{D}) \hat{J}^* \mathcal{B}^*, & \mathcal{C}^* \hat{J} \mathbf{D}(\mathcal{D}) & I \\ -\mathbf{D}(\mathcal{D}) \hat{J}^* \mathcal{B}^* & \mathbf{D}(\mathcal{D}) & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Theta(t) \\ \underline{v}(t) \\ \underline{u}_\Theta(t) \end{bmatrix} & \text{with } \underline{v}(t) = \begin{bmatrix} \underline{z}(t) \\ \underline{w}(t) \end{bmatrix}, \\
\mathbf{D}(\Theta) \begin{bmatrix} (K \star G)^\top \\ I \end{bmatrix} &:= \begin{cases} \begin{bmatrix} \mathcal{T}_G^* \dot{\mathbf{x}}_G(t) \\ \mathcal{T}^* \dot{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G^* + \mathcal{K}_G^* \mathcal{B}_u^* & 0 & \mathcal{C}_G^* + \mathcal{K}_G^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{B}}_1 \mathcal{B}_G^* + \mathcal{K}_\Theta^* \mathcal{B}_u^* & \underline{\mathcal{A}} & \underline{\mathcal{B}}_1 \mathcal{D}_G^* + \underline{\mathcal{B}}_2 + \mathcal{K}_\Theta^* \mathcal{D}_{zu}^* \\ \underline{\mathcal{D}}_{11} \mathcal{B}_G^* & \underline{\mathcal{C}}_1 & \underline{\mathcal{D}}_{11} \mathcal{D}_G^* + \underline{\mathcal{D}}_{12} \\ \underline{\mathcal{D}}_{21} \mathcal{B}_G^* & \underline{\mathcal{C}}_2 & \underline{\mathcal{D}}_{21} \mathcal{D}_G^* + \underline{\mathcal{D}}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ \underline{w}(t) \end{bmatrix} \end{cases} \quad (7)
\end{cases}
\end{aligned}$$

Figure 4: Dual components, realized using primal operators. $G^\top$ denotes the dual of G , $K^\top$ denotes the dual of K , $\mathbf{D}(\Theta)$ denotes the dual of Θ . $\mathbf{D}(\Theta) \begin{bmatrix} (K \star G)^\top \\ I \end{bmatrix}$ denotes the dual interconnection. Here $\mathbf{D}(\Theta) \begin{bmatrix} 0 \star G \\ I \end{bmatrix}$ denotes the dual interconnection with $\mathcal{K} = 0$.

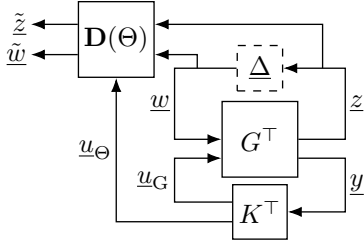


Figure 5: Dual controller interconnection.

7.1. Interpretation of the Dual System

Figure 5 is the dual representation of the primal closed loop in Figure 2. Duality maps $w \mapsto z$ to $\underline{w} \mapsto \underline{z}$, turns the state-feedback controller K into the dual feedback operator $K^\top$, which feeds $\underline{y}$ into the dual plant and filter, and replaces Θ by its dual $\mathbf{D}(\Theta)$. The block $\underline{\Delta}$ is only a formal dual of the uncertainty and need not be realized. Both representations use the zero initial condition in (A.2). If Θ is dynamic, controller recovery also uses (A.3).

7.2. State Feedback LPI

We now state the main synthesis condition.

Corollary 2. Let $K \star G$ be as in Definition 3, and let $(\Theta, J_{\mathbf{n}_z, \mathbf{n}_w})$ be a hard J -spectral factorization. Assume that both $K \star G$ and Θ admit PIE realizations depicted in (7). Let $\Delta \in \underline{\Delta}$ such that,

- The interconnection $K \star P \star \Delta$ is well-posed for all $\Delta \in \underline{\Delta}$.
- The IQC in Definition 5 is satisfied for all $\Delta \in \underline{\Delta}$.
- Let $\mathcal{D}$, $\mathcal{D}_G$ denote the feedthrough of Θ and G , such that $(\mathcal{D}_{21} \mathcal{D}_G + \mathcal{D}_{22})^{-1} \in \Pi_4$.

Moreover there exist $\epsilon > 0$ and $\mathcal{Z}, \mathcal{P} \in \Pi_4$ with $\mathcal{P} = \mathcal{P}^* \succ 0$, such that the following statement holds

$$\begin{aligned}
&\begin{bmatrix} \mathcal{T}^* \mathcal{P} \mathcal{A}_0 + \mathcal{T}^* \mathcal{Z} \begin{bmatrix} \mathcal{B}_u^* & 0 \end{bmatrix} + (*) & \mathcal{T}^* \mathcal{P} \mathcal{B}_0 + \mathcal{T}^* \mathcal{Z} \mathcal{D}_{zu}^* \\ (*) & \epsilon I \end{bmatrix} \\
&+ \begin{bmatrix} \underline{\mathcal{C}}^* \\ \underline{\mathcal{D}}^* \end{bmatrix} J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \preceq 0. \quad (10)
\end{aligned}$$

where

$$\mathbf{D}(\Theta) \begin{bmatrix} 0 \star G^\top \\ I \end{bmatrix} \underline{w} := \begin{cases} \begin{bmatrix} \mathcal{T} \underline{\xi}(t) \\ \underline{v}(t) \end{bmatrix} = \begin{bmatrix} \underline{\mathcal{A}}_0 & \underline{\mathcal{B}}_0 \\ \underline{\mathcal{C}} & \underline{\mathcal{D}} \end{bmatrix} \begin{bmatrix} \underline{\xi}(t) \\ \underline{w}(t) \end{bmatrix} \end{cases} \quad (11)$$

Let $\mathcal{K} = \mathcal{Z}^* \mathcal{P}^{-1}$, then the interconnection $K \star P \star \Delta$ with $\mathbf{x}(0) = 0$, as in (A.2), is robustly stable and robust performance is achieved.

Proof. See Appendix A.4. ■

The next subsections recover the controller using the reconstruction method in [25, Sec. VII], which analytically computes $\mathcal{K} = \mathcal{Z}^* \mathcal{P}^{-1}$ when $\mathcal{P} \succ 0$ is separable.

7.3. Memoryless Θ controller realization

When Θ is memoryless, no multiplier state is required in the controller. For a state-feedback operator $\mathcal{K}_G$, set $u(t) = \mathcal{K}_G \mathbf{x}_G(t)$. The resulting closed-loop PIE $K \star G$ is

$$\begin{bmatrix} \partial_t (\mathcal{T}_G \mathbf{x}_G)(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_G + \mathcal{B}_u \mathcal{K}_G & \mathcal{B}_G \\ \mathcal{C}_G + \mathcal{D}_{zu} \mathcal{K}_G & \mathcal{D}_G \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ w(t) \end{bmatrix}.$$

The controller is memoryless in time and acts directly on the measured or estimated PIE state.

7.4. Dynamic Θ controller realization

Dynamic-IQC feedback constructions with additional controller memory have also been studied for uncertain systems with state delay [43]. Under (A.3), for each dynamically filtered uncertainty channel, Δ is assumed to be known, so that z_Δ , $w_\Delta = \Delta z_\Delta$, and w_p are available for

filter-state reconstruction. Unknown uncertainty channels may be included simultaneously, provided they satisfy a memoryless hard IQC. Define the augmented state

$$\mathbf{x}(t) := \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \end{bmatrix}.$$

The recovered dynamic (nonlinear) controller is

$$K := \begin{cases} \mathcal{T}\dot{\mathbf{x}}_\Theta(t) = \mathcal{A}\mathbf{x}_\Theta(t) + \mathcal{B}^z z(t) + \mathcal{B}^w w(t), \\ u(t) = [\mathcal{K}_G \quad \mathcal{K}_\Theta] \mathbf{x}(t). \end{cases} \quad (12)$$

The numerical examples in the next section apply these memoryless and dynamic controller constructions and compare the resulting primal and dual certificates.

8. Numerical Examples

We consider PDE examples with repeated parametric and sector-bounded uncertainties. For additional information on the problem setup, the reader is referred to <https://github.com/TMLenssen/PIETOOLS-Duality>.

8.1. Note on loop shifting with $\mathcal{J}$

Define the spatial integration operator $\mathcal{J} \in \Pi_3$ by $(\mathcal{J}f)(t, s) := \int_0^s f(t, \theta) d\theta$. In our examples we perform the loop shift $\frac{\partial}{\partial s} \circ \Delta$ on distributed uncertainty channels entering the dynamics. The effect of this loop shift can be seen from $\dot{\mathbf{v}} = \mathbf{v}_{ss} + \delta^2 \mathbf{v}$, with $\mathbf{v}(t, 0) = 0$ and $|\delta| \leq \alpha$. Representing δ^2 by two repeated channels and spatially differentiating their inputs and outputs gives

$$\begin{aligned} \dot{\mathbf{v}}(t, s) &= \mathbf{v}_{ss}(t, s) + \mathcal{J}w_{\Delta,2}(t, s), \\ z_{\Delta,1}(t, s) &= \mathbf{v}_s(t, s), \quad z_{\Delta,2}(t, s) = w_{\Delta,1}(t, s), \\ w_{\Delta,i}(t, s) &= \delta z_{\Delta,i}(t, s), \quad i \in \{1, 2\}. \end{aligned}$$

Closing the loop recovers $\mathcal{J}w_{\Delta,2} = \delta^2 \mathcal{J}\mathbf{v}_s = \delta^2 \mathbf{v}$. The feedthrough $z_{\Delta,2} = w_{\Delta,1}$ remains unfiltered, since filtering it would instead produce the different term $\delta^2 \mathcal{J}\mathbf{v}$. This realization substantially improves feasibility in our computations, possibly because it better aligns with the fundamental PIE state. We do not have a general analytical explanation and therefore treat this benefit as empirical and realization- and uncertainty-dependent.

8.2. Primal–dual analysis with repeated real uncertainty

We begin with the finite-dimensional ODE benchmark in [46, Sec. 8.1]. We copy its temporal filter basis, and performance setup, and apply the same construction to a pair of coupled heat equations by lifting the filter over the spatial domain. For the static part we can use general PI operators. This multiplier setup aligns with the multiplier class in [32, Lem. 10]. We use the same finite-dimensional matrices, with the deviations declared below, as in the ODE benchmark, but apply them on distributed signals, and add a diffusion term to obtain the coupled PDE. Compared to [46, Sec. 8.1], A now carries the shift $\frac{\pi^2}{2}I$, which

offsets the diffusion eigenvalue, the performance input enters with the spatial weight s , as $sB_p w_p$, and the performance output carries a factor 2 on the uncertainty term, $z_p = \int_0^1 (C_p \mathbf{v} + 2D_{p\Delta} \mathcal{J}w_\Delta) ds$. Let $s \in [0, 1]$ and

$$\begin{aligned} A &= \begin{bmatrix} -2 & -3 \\ 1 & 1 \end{bmatrix} + \frac{\pi^2}{2}I, & B_\Delta &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, & B_p &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ C_\Delta &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, & D_\Delta &= \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}, & D_{\Delta p} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ C_p &= \begin{bmatrix} 1 & 0 \end{bmatrix}, & D_{p\Delta} &= \begin{bmatrix} 0 & 1 \end{bmatrix}. \end{aligned}$$

Consider

$$\begin{aligned} \dot{\mathbf{v}}(t, s) &= \mathbf{v}_{ss}(t, s) + A\mathbf{v}(t, s) + B_\Delta \mathcal{J}w_\Delta(t, s) + sB_p w_p(t), \\ z_\Delta(t, s) &= C_\Delta \mathbf{v}_s(t, s) + D_\Delta w_\Delta(t, s) + D_{\Delta p} w_p(t), \\ z_p(t) &= \int_0^1 (C_p \mathbf{v}(t, s) + 2D_{p\Delta} \mathcal{J}w_\Delta(t, s)) ds, \\ \mathbf{v}(t, 0) &= 0, \quad \mathbf{v}_s(t, 1) = 0, \\ w_\Delta(t, s) &= \delta z_\Delta(t, s), \quad |\delta| \leq \alpha, \end{aligned}$$

We convert this system to a PIE using the PIETOOLS toolbox [23]. We find that the uncertainty-channel PIE feedthrough $\mathcal{D}_\Delta$ of $w_\Delta \rightarrow z_\Delta$ acts pointwise:

$$\begin{aligned} (\mathcal{D}_\Delta \mathbf{q})(t, s) &= D_\Delta \mathbf{q}(t, s), \\ (\mathcal{D}_{\Delta\Delta} \mathbf{q})(t, s) &= \delta D_\Delta \mathbf{q}(t, s). \end{aligned}$$

Consequently, the algebraic well-posedness condition reduces to

$$\det(I_2 - \delta D_\Delta) = \det \begin{bmatrix} 1 - \delta & 2\delta \\ -\delta & 1 + \delta \end{bmatrix} = 1 + \delta^2 > 0.$$

Hence $I - \mathcal{D}_\Delta \Delta$ has a bounded inverse for every real δ . The same real scalar δ is repeated over both channels and all spatial locations, and $\Delta = \delta I$ is self-adjoint, bounded, time-invariant and memoryless. We therefore use the same multiplier family in the two analyses,

$$\begin{aligned} \mathcal{V}_\Delta \{\alpha^2 \underline{\mathcal{Q}}_\Delta, \alpha \underline{\mathcal{S}}_\Delta, -\underline{\mathcal{Q}}_\Delta\}, \quad \underline{\mathcal{Q}}_\Delta \succeq 0 \in \Pi_3, \quad \underline{\mathcal{S}}_\Delta^* = -\underline{\mathcal{S}}_\Delta \in \Pi_3, \\ \mathcal{V}_\Delta \{\alpha^2 \underline{\mathcal{Q}}_\Delta, \alpha \underline{\mathcal{S}}_\Delta, -\underline{\mathcal{Q}}_\Delta\}, \quad \underline{\mathcal{Q}}_\Delta \succeq 0 \in \Pi_3, \quad \underline{\mathcal{S}}_\Delta^* = -\underline{\mathcal{S}}_\Delta \in \Pi_3. \end{aligned}$$

This is the situation described in Section 6.1, because $\Delta = \Delta^*$, the primal and dual certificates can use the same dynamic filter and repeated-real IQC structure [32, Lem. 10]. The dynamic uncertainty filter is constructed from the stable temporal basis [46, Eq. (12a)],

$$\psi_\nu(i\omega) = \begin{bmatrix} 1 & \frac{1}{i\omega - \rho} & \cdots & \frac{1}{(i\omega - \rho)^\nu} \end{bmatrix}^\top, \quad \rho < 0,$$

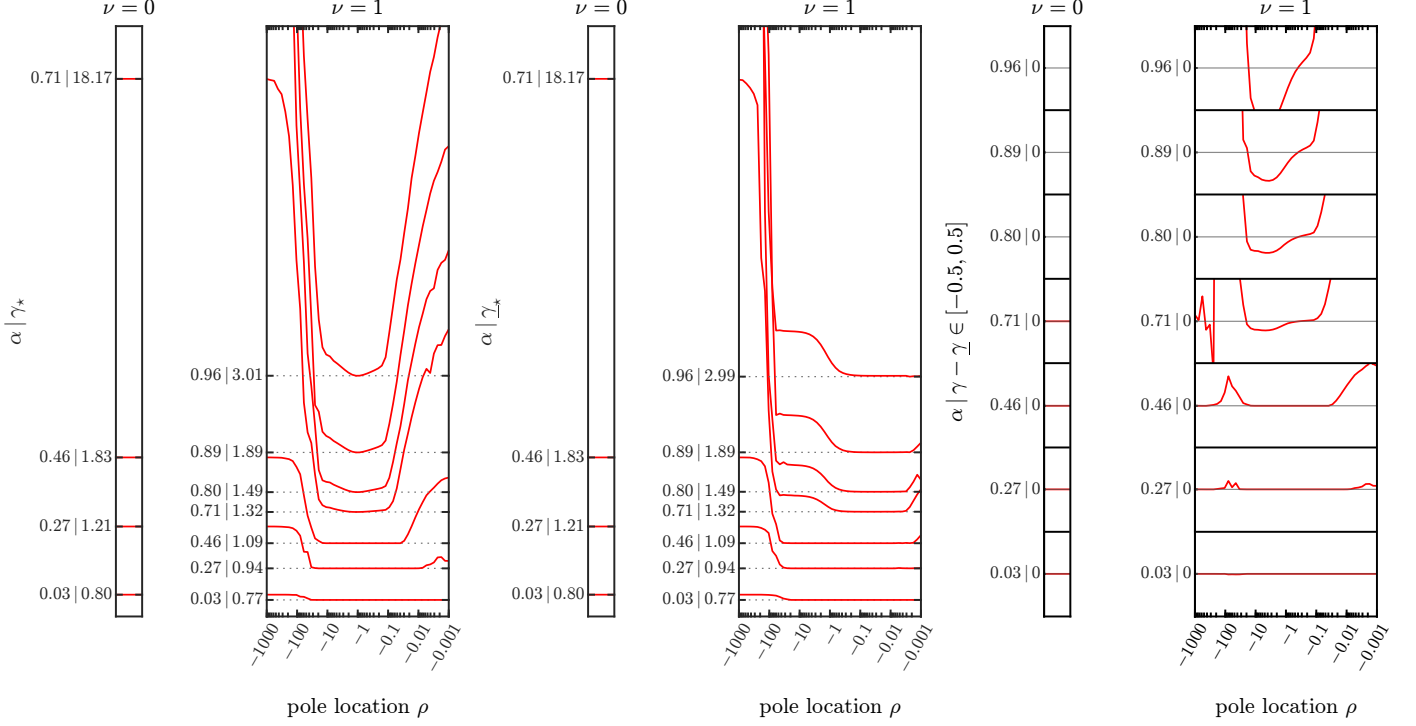


Figure 6: Reproduction of the sweep in [46, Fig. 7] for the coupled heat equations: primal analysis (left), dual analysis (center), and the difference $\gamma - \bar{\gamma}$ between the primal and dual bounds (right). The figure highlights two main observations. First, even when using a lifted finite-dimensional filter for the dynamic IQC, the choice $\nu = 1$ yields substantially smaller values of γ than the static case $\nu = 0$. Second, the dual analysis produces lower values of γ than the primal analysis. For this example and this parameterization the dual bound is the smaller one; the claim is not general, and Table 3 reports the opposite ordering for the reaction-diffusion plant, $\gamma = 0.5146$ against $\bar{\gamma} = 0.7432$. This difference is possible because the problem is formulated such that Θ need not be $\mathbf{D}(\Theta)$, so the primal and dual analyses need not yield identical bounds under the same lifted finite-dimensional filter parameterization, which may be better suited to the dual system.

For $\nu \geq 1$, define the finite-dimensional realization

$$A_\nu = \begin{bmatrix} \rho & 0 & \cdots & 0 \\ 1 & \rho & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 1 & \rho \end{bmatrix} \in \mathbb{R}^{\nu \times \nu}, \quad B_\nu = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{\nu \times 1},$$

$$C_\nu = \begin{bmatrix} 0_{1 \times \nu} \\ I_\nu \end{bmatrix} \in \mathbb{R}^{(1+\nu) \times \nu},$$

$$D_\nu = \begin{bmatrix} 1 \\ 0_{\nu \times 1} \end{bmatrix} \in \mathbb{R}^{(1+\nu) \times 1}.$$

Since there are two uncertainty channels, define the PI operators pointwise by

$$\begin{aligned} (\mathcal{T}\mathbf{q})(s) &= \mathbf{q}(s), \\ (\mathcal{A}\mathbf{q})(s) &= (A_\nu \otimes I_2)\mathbf{q}(s), \quad (\mathcal{B}\mathbf{q})(s) = (B_\nu \otimes I_2)\mathbf{q}(s), \\ (\mathcal{C}\mathbf{q})(s) &= (C_\nu \otimes I_2)\mathbf{q}(s), \quad (\mathcal{D}\mathbf{q})(s) = (D_\nu \otimes I_2)\mathbf{q}(s). \end{aligned}$$

Thus the lifted basis function satisfies $\psi\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$. The uncertainty filter used in both analyses is

$$\Psi_\Delta = \Psi_\Delta := \begin{bmatrix} \psi & 0 \\ 0 & \psi \end{bmatrix}, \quad \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}. \quad (13)$$

In particular, no distinct dual filter is introduced in this example. The gain from w_p to z_p is tested with the induced L_2 -gain performance multipliers $\mathcal{V}_p\{1, 0, -\gamma^2\}$ and

Table 1: One-shot solver computation time at $\alpha = 0.5$ and $\rho = -1$.

Filter order ν	Primal (sec)	Dual (sec)
0	0.74	0.70
1	15.61	9.98
2	82.34	56.82
3	327.82	194.86

$\mathcal{V}_p\{1, 0, -\bar{\gamma}^2\}$, respectively. Table 1 shows that computation time increases sharply with filter order. We therefore perform the full pole sweep in Figure 6 only for $\nu = 1$, with $\nu = 0$ included as the static baseline. The figure shows the optimized gains and their difference.

8.3. Boundary control with sector-bounded uncertainty

We synthesize boundary controllers for a nonlinear reaction-diffusion equation and a nonlinear damped wave equation. Related finite-horizon control of the sine-Gordon equation is studied in [49]. Index the reaction-diffusion and wave equations by $i = 1$ and $i = 2$, respectively, with distributed state $\mathbf{v}_i$ and boundary value x_i . On $s \in [0, 1]$, the two plants and their common boundary

conditions, for $i \in \{1, 2\}$, are

$$\begin{aligned}\dot{\mathbf{v}}_1(t, s) &= \mathbf{v}_{1,ss}(t, s) + 5 \sin(\mathbf{v}_1(t, s)) + sw_p(t), \\ \ddot{\mathbf{v}}_2(t, s) &= \mathbf{v}_{2,ss}(t, s) - 2\dot{\mathbf{v}}_2(t, s) + 3 \sin(\mathbf{v}_2(t, s)) + sw_p(t), \\ \mathbf{v}_i(t, 0) &= 0, \quad \mathbf{v}_{i,s}(t, 1) = x_i(t), \\ \dot{x}_i(t) &= u_i(t), \quad z_{p,i}(t) = \left[\int_0^1 \begin{matrix} x_i(t) \\ \mathbf{v}_i(t, s) \end{matrix} ds \right].\end{aligned}$$

Both plants are converted to a PIE using the PIETOOLS library [23]. The boundary value is a controller state and its derivative is the control input, (A.4). Using $\mathcal{J}$ from the preceding subsection, set $z_{\Delta,i} = \mathbf{v}_{i,s}$ and $w_{\Delta,i} = \cos(\mathcal{J}z_{\Delta,i})z_{\Delta,i}$. Since $\mathcal{J}z_{\Delta,i} = \mathbf{v}_i$ and $\mathbf{v}_i(t, 0) = 0$, integration gives $\mathcal{J}w_{\Delta,i} = \sin(\mathbf{v}_i)$. Thus $\lambda_i \sin(\mathbf{v}_i)$ is replaced exactly by $\lambda_i \mathcal{J}w_{\Delta,i}$, where $\lambda_1 = 5$ and $\lambda_2 = 3$, while $sw_p(t) = (\mathcal{J}w_p)(t, s)$. The sector bound $-z_{\Delta,i}^2 \leq z_{\Delta,i}w_{\Delta,i} \leq z_{\Delta,i}^2$ holds for both plants, so the same $[-1, 1]$ sector IQC applies.

Lemma 1 (Dual sector-bound IQC). *Suppose that $(\Delta z_{\Delta})(t, s) = \phi(z_{\Delta}(t, s))$ and*

$$\alpha z^2 \leq z\phi(z) \leq \beta z^2, \quad z \in \mathbb{R},$$

where $\alpha < \beta$. Then Δ satisfies the dual hard IQC with $\underline{\Psi}_{\Delta} = I$ and

$$\underline{\mathcal{V}}_{\Delta} = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{\mathcal{S}}_{\Delta} \\ \underline{\mathcal{S}}_{\Delta} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}, \quad (14)$$

where $\underline{\mathcal{S}}_{\Delta} \in \Pi_3$ is the positive multiplication operator

$$(\underline{\mathcal{S}}_{\Delta} \mathbf{x})(t, s) = \underline{\mathcal{S}}_0(s) \mathbf{x}(t, s), \quad \underline{\mathcal{S}}_0(s) > 0. \quad (15)$$

Proof. See Appendix A.5. $\blacksquare$

With $\mathcal{S}_{\Delta} > 0$ denoting the primal sector weight, the factorization in Appendix A.5 gives the invertible well-posedness condition

$$\mathcal{D}_{21,\Delta} \mathcal{D}_{P,\Delta} + \mathcal{D}_{22,\Delta} = \frac{(I - \mathcal{S}_{\Delta}) \mathcal{D}_{P,\Delta} + I + \mathcal{S}_{\Delta}}{\sqrt{2}}. \quad (16)$$

Since S_0 is a polynomial, positive and continuous on the compact $[a, b]$, there are $0 < m \leq S_0 \leq M$, so $(I + \mathcal{S}_{\Delta})^{-1}(I - \mathcal{S}_{\Delta})$ is multiplication by $(1 - S_0(s))/(1 + S_0(s))$ and has norm $\kappa = \max\{(1 - m)/(1 + m), (M - 1)/(M + 1)\} < 1$. Hence $\|\mathcal{D}_{P,\Delta}\| \leq 1$ makes $\mathcal{D}_{21,\Delta} \mathcal{D}_{P,\Delta} + \mathcal{D}_{22,\Delta}$ boundedly invertible, and $\mathcal{D}_{P,\Delta} = 0$ is the special case. We use Lemma 1 with $\alpha = -1$ and $\beta = 1$, together with the induced L_2 -gain performance multiplier $\underline{\mathcal{V}}_p\{1, 0, -\gamma_i^2 I_2\}$, and $\|\mathcal{K}_i\| \leq 10^4$. The recovered controllers act on the PIE states $[x_1 \quad \mathbf{v}_{1,ss}]^T$ for reaction–diffusion and $[x_2 \quad \dot{\mathbf{v}}_2 \quad \mathbf{v}_{2,ss}]^T$ for the wave equation. Thus, in PDE coordinates, their respective feedback laws are

$$\begin{aligned}u_1(t) &= k_1 x_1(t) + \int_0^1 Q_1(s) \mathbf{v}_{1,ss}(t, s) ds, \\ u_2(t) &= k_2 x_2(t) + \int_0^1 \begin{bmatrix} Q_{2,1}(s) & Q_{2,2}(s) \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}}_2(t, s) \\ \mathbf{v}_{2,ss}(t, s) \end{bmatrix} ds.\end{aligned}$$

Table 2: Controller coefficients for the reaction–diffusion and wave feedback laws, rounded to two decimal places. $[s^n]Q$ denotes the coefficient of s^n in $Q(s)$.

n	Reaction–diffusion	Wave	
	$[s^n]Q_1$	$[s^n]Q_{2,1}$	$[s^n]Q_{2,2}$
0	1.74	−8.06	−0.04
1	−453.64	1.58	11.37
2	120.33	−27.33	50.18
3	−767.56	−7.38	−104.87
4	1543.37	50.41	13.91
5	−2646.43	−88.60	−151.93
6	2810.30	97.81	−218.07
7	−983.50	−115.17	129.53
8	−1505.86	117.26	−8.59
9	1921.83	−54.48	−11.42
10	−652.91	7.28	−0.77
Gain	$k_1 = -331.62$	$k_2 = -37.71$	

Table 3: Synthesis and analysis gain bounds, and simulated open- and closed-loop L_2 ratios over $0 \leq t \leq 35$.

Quantity	Reaction–diffusion	Wave
Synthesized γ_s	1.268	−
Primal Analysis γ	0.5146	−
Dual Analysis $\underline{\gamma}$	0.7432	1.373
Open-loop L_2 ratio	0.4871	0.3152
Closed-loop L_2 ratio	0.4605	0.2624

The dual synthesis LPI produces stabilizing boundary controllers for both plants, whose coefficients are listed in Table 2. Figure 7 compares their open- and closed-loop responses to the exogenous disturbance $w_p(t) = 8 \cos(t) \mathbf{1}_{[\pi, 4\pi]}(t)$. Table 3 reports the synthesis gain bound γ_s , the primal and dual analysis bounds γ and $\underline{\gamma}$, and the simulated open- and closed-loop ratios $\|z_p\|_{L_2[0,35]} / \|w_p\|_{L_2[0,35]}$. Figure 8 shows the corresponding control effort. Controller synthesis on the wave equation yielded an infeasible LPI, but returned a controller which was validated using the dual analysis LPI. The closed-loop ratios differ little from the open-loop ones, 0.4871 against 0.4605 and 0.3152 against 0.2624. The reason is that in open-loop simulation there is no boundary control effort, i.e., $x_i(t) = 0$; thus simulating over a longer time horizon would display a stronger difference in L_2 -gain.

9. Connection to finite-dimensional dynamic multipliers

We relate the duality framework to finite-dimensional IQC multipliers and apply the resulting synthesis conditions to a delayed neural population model.

9.1. Positive–negative multipliers

Let $\mathbb{R}H_{\infty}$ denote the proper real-rational transfer matrices with no poles in the closed right half-plane. For

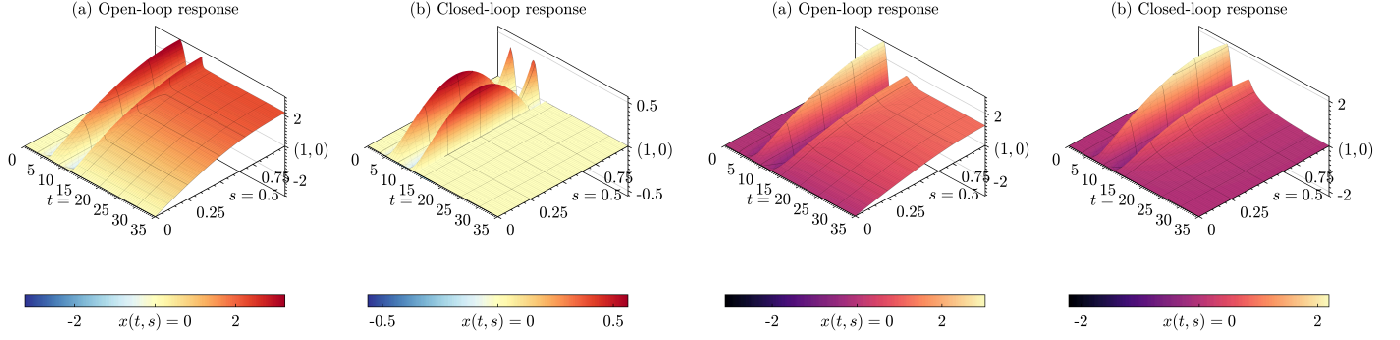


Figure 7: Open- and closed-loop state responses for the nonlinear reaction–diffusion equation (left pair) and the damped wave equation (right pair). Disturbed by $w_p(t) = 8 \cos(t) \mathbf{1}_{[\pi, 4\pi]}(t)$, we find that the closed-loop response drives the system towards equilibrium.

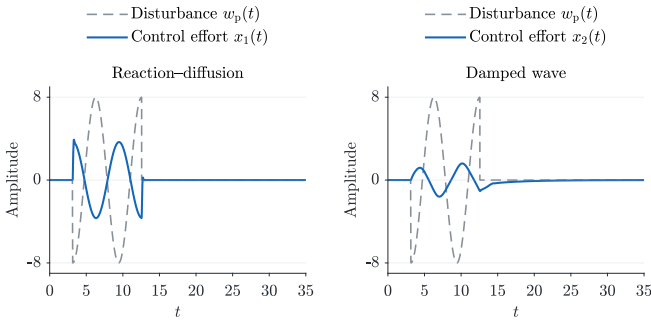


Figure 8: Disturbance and control effort for the reaction–diffusion equation (left) and damped wave equation (right).

$\Psi \in \mathbb{R}H_\infty^{(n_z+n_w) \times (n_z+n_w)}$, define the para-Hermitian adjoint by

$$\Psi^\sim(i\omega) = \Psi(-i\omega)^\top \in \mathbb{C}^{(n_z+n_w) \times (n_z+n_w)}.$$

Given a constant symmetric matrix $V \in \mathbb{R}^{(n_z+n_w) \times (n_z+n_w)}$, partition $\Pi = \Psi^\sim V \Psi$ according to the input dimensions n_z and n_w , so that $\Pi(i\omega) \in \mathbb{C}^{(n_z+n_w) \times (n_z+n_w)}$,

We say that Π is strict positive–negative (PN) if there exists $\epsilon > 0$ such that

$$\Pi(i\omega) = \begin{bmatrix} \Pi_{11}(i\omega) & \Pi_{12}(i\omega) \\ \Pi_{12}(i\omega)^* & \Pi_{22}(i\omega) \end{bmatrix},$$

$$\Pi_{11}(i\omega) \succeq \epsilon I_{n_z}, \quad \Pi_{22}(i\omega) \preceq -\epsilon I_{n_w}$$

for all $\omega \in \mathbb{R} \cup \{\infty\}$. By [40, Thm. 4], a strict PN multiplier admits a square J -spectral factor

$$\Pi = \Theta^\sim J \Theta, \quad J = \text{diag}(I_{n_z}, -I_{n_w}),$$

with $J \in \mathbb{R}^{(n_z+n_w) \times (n_z+n_w)}$ and $\Theta, \Theta^{-1} \in \mathbb{R}H_\infty^{(n_z+n_w) \times (n_z+n_w)}$. Note that [40, Thm. 4] asserts $\Theta, \Theta^{-1} \in \mathbb{R}H_\infty$ and hardness, but nothing about the (2,2) block. What the construction below does deliver is nonsingularity, $\Theta_{22}^\sim \Theta_{22} \succeq \epsilon I$ on $i\mathbb{R} \cup \{\infty\}$. Accordingly, $\Theta_{22}^{-1} \in \mathbb{R}H_\infty^{n_w \times n_w}$ is imposed as the second condition of Definition 7, and the dual filter is

$\mathbf{D}(\Theta) \in \mathbb{R}H_\infty^{(n_w+n_z) \times (n_w+n_z)}$. The Riccati construction in [50, App. B] uses the factorization results of [39] and provides a computational method to find factorizations. Related finite-horizon formulations are given in [41, 42].

9.2. Robust stabilization of the STN–GPe network

We consider the delayed firing-rate model of [51, Eq. (5)] with $K_d = 1$, i.e. the diseased case. Let STN and GP denote the firing rates of the subthalamic nucleus (STN) and globus pallidus pars externa (GPe), respectively, and replace $I_e \text{Stim}$ with $u(t)$. The plant is

$$\begin{aligned} \tau_S \dot{\text{STN}}(t) &= F_S(-w_{\text{GS}} \text{GP}(t - \tau_{\text{GS}}) \\ &\quad + w_{\text{CS}} \text{Ctx}) - \text{STN}(t) + u(t), \\ \tau_G \dot{\text{GP}}(t) &= F_G(w_{\text{SG}} \text{STN}(t - \tau_{\text{SG}}) \\ &\quad - w_{\text{GG}} \text{GP}(t - \tau_{\text{GG}}) \\ &\quad - w_{\text{XG}} \text{Str}) - \text{GP}(t). \end{aligned} \quad (17)$$

Table 4: STN–GPe model parameters for the diseased case ($K_d = 1$), from [51, Tables 1 and 3]. The transmission delays denoted by Δt_{ij} in the source are written as τ_{ij} here.

Parameter	Value	Unit
K_d	1	–
w_{SG}	20.0	–
w_{GS}	10.7	–
w_{GG}	12.3	–
w_{CS}	9.2	–
w_{XG}	139.4	–
τ_S	6	ms
τ_G	14	ms
τ_{SG}	6	ms
τ_{GS}	6	ms
τ_{GG}	4	ms
Ctx	27	spikes/s
Str	2	spikes/s
M_S	300	spikes/s
M_G	400	spikes/s
B_S	17	spikes/s
B_G	75	spikes/s

The parameter values for the diseased case are listed in Table 4. The activation functions $F_i: \mathbb{R} \rightarrow \mathbb{R}$ are

$$F_i(q) = \frac{M_i}{1 + \exp(-4q/M_i)(M_i - B_i)/B_i)}, \quad i \in \{S, G\}, \quad (18)$$

with $F_i(0) = B_i$. Define the sigmoid midpoint and centered activation $\sigma_i: \mathbb{R} \rightarrow \mathbb{R}$ by

$$c_i := \frac{M_i}{4} \ln\left(\frac{M_i - B_i}{B_i}\right),$$

$$\sigma_i(q) := \frac{M_i}{1 + \exp(-4q/M_i)} - \frac{M_i}{2} = \frac{M_i}{2} \tanh\left(\frac{2q}{M_i}\right). \quad (19)$$

Then $F_i(q) = \sigma_i(q - c_i) + M_i/2$. With the centered states

$$\tilde{x}_S := \text{STN} - \frac{M_S}{2}, \quad \tilde{x}_G := \text{GP} - \frac{M_G}{2},$$

and offsets

$$b_S := w_{CS}\text{Ctx} - \frac{w_{GS}M_G}{2} - c_S,$$

$$b_G := \frac{w_{SG}M_S}{2} - \frac{w_{GG}M_G}{2} - w_{XG}\text{Str} - c_G,$$

the plant becomes

$$\begin{aligned} \dot{\tilde{z}}_S(t) &:= -w_{GS}\tilde{x}_G(t - \tau_{GS}) + b_S, \\ \dot{\tilde{z}}_G(t) &:= w_{SG}\tilde{x}_S(t - \tau_{SG}) - w_{GG}\tilde{x}_G(t - \tau_{GG}) + b_G, \\ \tau_S \dot{\tilde{x}}_S(t) &= \sigma_S(\tilde{z}_S(t)) - \tilde{x}_S(t) + \tilde{u}(t), \\ \tau_G \dot{\tilde{x}}_G(t) &= \sigma_G(\tilde{z}_G(t)) - \tilde{x}_G(t). \end{aligned} \quad (20)$$

where $\sigma_i(0) = 0$. Here τ_{ij} and w_{ij} denote the transmission delay and synaptic weight from population i to population j , while τ_S and τ_G are the population time constants. Let $(\tilde{x}_S^*, \tilde{x}_G^*)$ be an equilibrium satisfying

$$\begin{aligned} \tilde{z}_S^* &:= -w_{GS}\tilde{x}_G^* + b_S, \\ \tilde{z}_G^* &:= w_{SG}\tilde{x}_S^* - w_{GG}\tilde{x}_G^* + b_G, \\ \tilde{x}_S^* &= \sigma_S(\tilde{z}_S^*), \\ \tilde{x}_G^* &= \sigma_G(\tilde{z}_G^*). \end{aligned}$$

For $i \in \{S, G\}$, introduce the deviation variables

$$x_i(t) := \tilde{x}_i(t) - \tilde{x}_i^*, \quad z_i(t) := \tilde{z}_i(t) - \tilde{z}_i^*,$$

and define the shifted nonlinearities $\delta_i: \mathbb{R} \rightarrow \mathbb{R}$ by

$$\delta_i(v) := \sigma_i(\tilde{z}_i^* + v) - \sigma_i(\tilde{z}_i^*).$$

Following [52], introduce the delayed states $\phi_{ij}(t, s) := x_i(t - \tau_{ij}s)$ for $(i, j) \in \mathcal{I}_D$ and $s \in [0, 1]$, where $\mathcal{I}_D := \{(G, S), (S, G), (G, G)\}$. The shifted plant has the equivalent ODE-PDE representation

$$\begin{aligned} \dot{z}_S(t) &:= -w_{GS}\phi_{GS}(t, 1), \\ \dot{z}_G(t) &:= w_{SG}\phi_{SG}(t, 1) - w_{GG}\phi_{GG}(t, 1), \\ \tau_S \dot{x}_S(t) &= \delta_S(z_S(t)) - x_S(t) + u(t), \\ \tau_G \dot{x}_G(t) &= \delta_G(z_G(t)) - x_G(t), \\ \dot{\phi}_{ij}(t, s) &= -\frac{1}{\tau_{ij}}\phi_{ij,s}(t, s), \\ \phi_{ij}(t, 0) &= x_i(t), \quad (i, j) \in \mathcal{I}_D. \end{aligned} \quad (21)$$

The equilibrium is now at the origin, and the shift affects only the nonlinearities δ_i . We synthesize feedback through u to robustly stabilize this equilibrium.

9.3. Robust stability

The diagonal uncertainty acting on two-dimensional signals is $\Delta = \text{diag}(\delta_S, \delta_G)$. It is memoryless, monotone, and zero at the origin. We compare a static sector IQC with a Zames-Falb IQC [29, 53]. For each channel, let $M, H \in \mathbb{R}H_\infty$ satisfy $M = 1 - H$ with

$$H(i\omega) = \sum_{k=1}^3 \frac{\kappa_k}{i\omega + a_k}, \quad a_k > 0, \quad \kappa_k \geq 0.$$

The nonnegative impulse response h satisfies $\|h\|_{L_1} = \sum_k \kappa_k/a_k < 1$, so the multiplier is valid for monotone slope-restricted nonlinearities without requiring oddness. We use $\tau_S = \tau_{GS} = \tau_{SG} = 6$ ms and $\tau_G = 14$ ms and the recurrent GPe delay $\tau_{GG} = 4$ ms. Specifically,

$$(a_1, a_2, a_3) = \left(\frac{1}{\tau_G}, \frac{1}{\tau_S}, \frac{1}{\tau_{GG}}\right)$$

Thus the multiplier poles are $-a_k$. The three modes are weighted by the corresponding synaptic strengths, which at $K_d = 1$ are $w_{GS} = 10.7$, $w_{SG} = 20.0$ and $w_{GG} = 12.3$.

$$(r_1, r_2, r_3) := \frac{(w_{GS}, w_{SG}, w_{GG})}{w_{GS} + w_{SG} + w_{GG}}$$

$$\kappa_k := \frac{10}{15} r_k a_k.$$

This gives $\|h\|_{L_1} = \sum_k \kappa_k/a_k = 10/15 < 1$. The time scales and synaptic weights specify a fixed heuristic multiplier shape, its coefficients are held fixed during the search over α . For each candidate bound $0 < \alpha \leq 1$, the multiplier $\Pi_{ZF}(i\omega) \in \mathbb{C}^{2 \times 2}$ before regularization is

$$\Pi_{ZF}(i\omega) = \begin{bmatrix} 0 & \alpha M(i\omega)^\sim \\ \alpha M(i\omega) & -(M(i\omega) + M(i\omega)^\sim) \end{bmatrix}.$$

Since the upper-left entry is zero, Π_{ZF} is not strict PN. Both uncertainty classes satisfy $|\delta_i(v)| \leq |v|$ and hence the IQC can be regularized with weight $\begin{bmatrix} \epsilon & 0 \\ 0 & -\epsilon \end{bmatrix}$ for some $\epsilon > 0$

$$\Pi_\epsilon = \Pi_{ZF} + \begin{bmatrix} \epsilon & 0 \\ 0 & -\epsilon \end{bmatrix},$$

the resulting multiplier remains a valid IQC and satisfies the strict PN requirement. Thus the multiplier and admits the hard J -spectral factor used in the dual-filter construction. The static sector case follows from Lemma 1.

9.3.1. Admissible trajectories

In this section we perform controller synthesis, as in Corollary 2, to find for which trajectories the system remains robustly stable. Bisection searches applying the

same sector and slope restrictions on δ_S and δ_G yield the certified bounds

$$\alpha_{\text{sec}}^* = 0.492, \quad \alpha_{\text{ZF}}^* = 0.562.$$

For $i \in \{S, G\}$, the sector condition on an interval I_i is

$$0 \leq \frac{\delta_i(v)}{v} \leq \alpha_{\text{sec}}^*, \quad v \in I_{\text{sec},i} \setminus \{0\}.$$

The Zames–Falb multiplier uses the slope condition

$$0 \leq \frac{\delta_i(v) - \delta_i(w)}{v - w} \leq \alpha_{\text{ZF}}^*, \quad v, w \in I_{\text{slope},i}, \quad v \neq w.$$

For each constraint, the maximal interval containing zero determines the admissible trajectories z_S, z_G corresponding to the $K_d = 1$ equilibrium is shown in Figure 9.

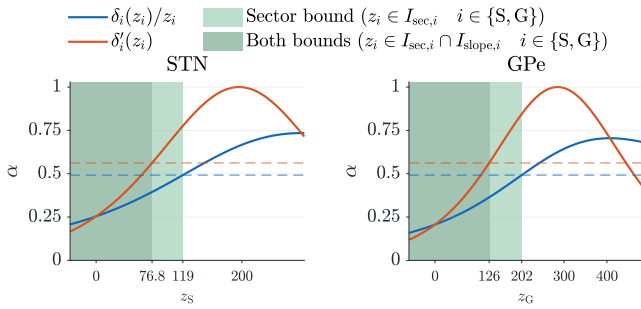


Figure 9: Sector bounds (blue) and slope bounds (orange) for the shifted STN and GPe nonlinearities. The dashed horizontal lines indicate the maximum certified values $\alpha_{\text{sec}}^* = 0.492$ and $\alpha_{\text{ZF}}^* = 0.562$ obtained using the sector and Zames–Falb multipliers, respectively. The shaded regions indicate the ranges of z_G and z_S for which robust stability can be guaranteed. Although the Zames–Falb multiplier certifies a larger value of α , i.e., $\alpha_{\text{sec}}^* < \alpha_{\text{ZF}}^*$, the corresponding admissible trajectory interval is smaller. This can be seen from the sector-bounded regions in the figure, including the region in which both bounds are satisfied.

9.3.2. Closed-loop simulation

Figure 10 shows the shifted nonlinear response with the controller synthesized using only the static sector IQC and the robust-stability condition, at the certified value $\alpha_{\text{sec}}^* = 0.492$. The initial condition is $x_S(0) = x_G(0) = 0$. To perturb the $K \star P \star \Delta$ interconnection, we add a pulse in interconnection (2), $w_\Delta = \Delta z_\Delta + d$, where

$$d(t) = \begin{cases} \begin{bmatrix} 10 \\ -10 \end{bmatrix} \sin^2\left(\frac{\pi(t-0.10)}{0.04}\right), & 0.10 \leq t \leq 0.14, \\ 0, & \text{otherwise.} \end{cases}$$

The horizontal lines mark the sector limits $z_i^{\max}$, defined by the first positive crossing of $\delta_i(z_i)/z_i = 0.492$ as depicted in Figure 9.

10. Conclusion

This paper established primal–dual IQC equivalence for PIE systems through J -spectral factorization and the induced L_2 -gain duality result. The dual-filter construction

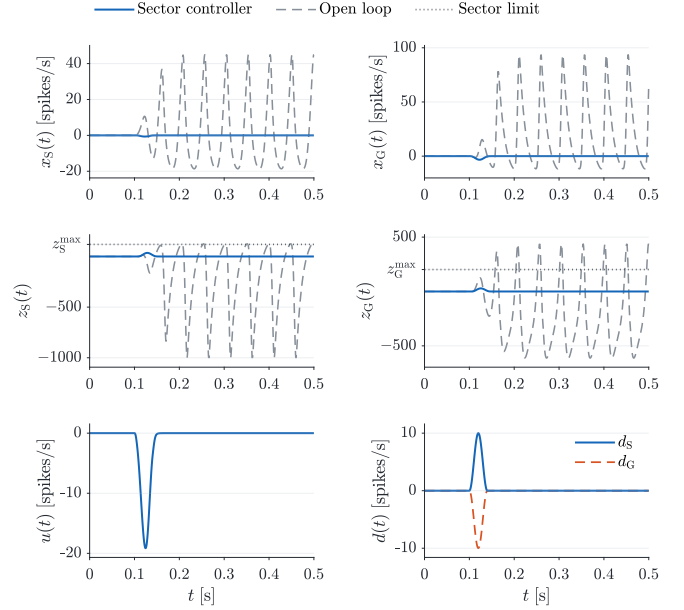


Figure 10: Shifted nonlinear STN–GPe simulation with the controller synthesized using the sector IQC. The top panels show the firing-rate deviations of the STN and GPe populations. The middle panels show the shifted nonlinearity inputs, $z_S(t)$ and $z_G(t)$, together with their sector-admissible upper limits, $z_S^{\max} = 119.09$ and $z_G^{\max} = 201.85$. Both inputs remain within the certified closed-loop robust-stability region throughout the simulation. The bottom panels show the control effort and the injected disturbance, respectively. The disturbance (bottom right) consists of $d_S(t)$ (blue) and $d_G(t)$ (dashed orange).

gives equivalent strict primal and dual IQC inequalities and a dual robust-performance certificate. It supports infinite-dimensional channels for which a hard J -spectral factorization is available, one of which is constructed here for the sector class, and memoryless or dynamic filters.

As an application, the dual robust-performance certificate yields a convex LPI condition for robust state-feedback synthesis. The resulting closed-loop interconnection is robustly stable and achieves the prescribed robust performance. This extends nominal PIE state-feedback synthesis beyond unstructured small-gain conditions. The numerical examples illustrate the primal and dual analysis conditions and the state-feedback synthesis condition. They also show that computation time increases sharply with filter order, motivating efficient multiplier selection.

The dual IQC representation also paves the way for robust estimator and feedforward synthesis, as in the finite-dimensional setting [44, 45]. Developing an elimination lemma for PIE systems would allow this approach to be investigated for infinite-dimensional systems. Further work also includes hard J -spectral factorizations for multipliers acting on infinite-dimensional channels.

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Appendix A. Proofs

Appendix A.1. Proof of Theorem 1, Veenman et al. [46]

Suppose that (3) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{L[a,b]} + \left\langle p_T \Psi_p \begin{bmatrix} P_p \Delta \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} P_p \Delta \\ 0 \end{bmatrix} w_\Delta \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_\Delta\|_{L[a,b]}^2.$$

By Definition 6 we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_\Delta\|_{L[a,b]}^2.$$

Then, using Theorem 6 in [32], we find that *robust stability* is guaranteed. Moreover, rewriting (3) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{L[a,b]} + \left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon (\|p_T w_\Delta\|_{L[a,b]}^2 + \|p_T w_p\|_{L[a,b]}^2),$$

and given that Ψ_Δ and $\mathcal{M}_{\mathcal{V}_\Delta}$ satisfy the IQC in Definition 5 leads to,

$$\left\langle p_T \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \Psi_p \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{L[a,b]} \leq -\epsilon \|p_T w_p\|_{L[a,b]}^2.$$

Thus by Definition 6 *robust performance* is achieved.

Appendix A.2. Proof of Theorem 3

1) $\implies$ 2) Assume there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for all $\tilde{w} \in L[a,b]$ and $T \geq 0$. We need to show that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L[a,b]}^2$ for some $\epsilon > 0$. First since $\Gamma := \Theta_{21}P + \Theta_{22}$ is causally boundedly invertible, $\Theta \begin{bmatrix} P \\ I \end{bmatrix} w = \begin{bmatrix} G_0 \\ I \end{bmatrix} \Gamma w$, and hence $\mathcal{I} = \|p_T G_0 \Gamma w\|_{L[a,b]}^2 - \|p_T \Gamma w\|_{L[a,b]}^2$. By statement 1 and causality of G_0 ,

$$\mathcal{I} \leq -(1 - \rho^2) \|p_T \Gamma w\|_{L[a,b]}^2 \leq -\frac{1 - \rho^2}{\|\Gamma^{-1}\|^2} \|p_T w\|_{L[a,b]}^2,$$

where the second inequality follows from $\|p_T w\|_{L[a,b]} = \|p_T \Gamma^{-1} p_T \Gamma w\|_{L[a,b]} \leq \|\Gamma^{-1}\| \|p_T \Gamma w\|_{L[a,b]}$. Thus, $\epsilon = \frac{(1 - \rho^2)}{\|\Gamma^{-1}\|^2} > 0$.

2) $\implies$ 1) Assume that there exists a $\epsilon > 0$ such that $\mathcal{I} \leq -\epsilon \|p_T w\|_{L[a,b]}^2$ for every w , and every $T \geq 0$. We need to establish there exists a $\rho < 1$ such that $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for all $\tilde{w} \in L[a,b]$ and $T \geq 0$. For arbitrary $\tilde{w}$, set $w = \Gamma^{-1} \tilde{w}$. Then

$$\|p_T G_0 \tilde{w}\|_{L[a,b]}^2 - \|p_T \tilde{w}\|_{L[a,b]}^2 \leq -\epsilon \|p_T \Gamma^{-1} \tilde{w}\|_{L[a,b]}^2.$$

Since Γ is bounded and causal, $\|p_T \tilde{w}\|_{L[a,b]} \leq \|\Gamma\| \|p_T \Gamma^{-1} \tilde{w}\|_{L[a,b]}$. Therefore,

$$\|p_T G_0 \tilde{w}\|_{L[a,b]}^2 \leq \left(1 - \frac{\epsilon}{\|\Gamma\|^2}\right) \|p_T \tilde{w}\|_{L[a,b]}^2.$$

Choose $\rho < 1$ such that $1 - \epsilon/\|\Gamma\|^2 \leq \rho^2 < 1$, then $\|p_T G_0 \tilde{w}\|_{L[a,b]} \leq \rho \|p_T \tilde{w}\|_{L[a,b]}$ for every $\tilde{w}$ and $T \geq 0$.

1) $\iff$ 3) We show that there exists a causally boundedly invertible $\underline{\Gamma}$ such that

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} \underline{w}.$$

then we can apply the previous arguments to show 1) $\iff$ 3). Define

$$S := \Theta_{11} - \Theta_{12} \Theta_{22}^{-1} \Theta_{21}.$$

Since Θ and Θ_{22} are boundedly invertible, so is S by Schur complement. Block inversion of $\mathbf{D}(\Theta)$ gives

$$\mathbf{D}(\Theta)_{21} = (S^\top)^{-1} \Theta_{21}^\top (\Theta_{22}^\top)^{-1}, \quad \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}.$$

Therefore, define

$$\underline{\Gamma} := \mathbf{D}(\Theta)_{21}P^\top + \mathbf{D}(\Theta)_{22} = (S^\top)^{-1}(I + P\Theta_{22}^{-1}\Theta_{21})^\top.$$

Now $\Gamma = \Theta_{22}(I + \Theta_{22}^{-1}\Theta_{21}P)$. Thus, invertibility of Γ and the push-through rule imply that $I + P\Theta_{22}^{-1}\Theta_{21}$, and hence $\underline{\Gamma}$, is causally boundedly invertible. Normalizing the upper block gives

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} \underline{G}_0 \\ I \end{bmatrix} \underline{\Gamma} w, \quad \underline{G}_0 := (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top).$$

Using

$$\underline{\Gamma}^{-\top} = S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}, \quad (I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} = P\Gamma^{-1},$$

where the second identity is the push-through rule, yields

$$\begin{aligned} \underline{G}_0 &= (\Theta_{22}^\top)^{-1}(P^\top \underline{\Gamma}^{-1} + \Theta_{12}^\top) \\ &= \left(\underline{\Gamma}^{-\top} P\Theta_{22}^{-1} + \Theta_{12}\Theta_{22}^{-1} \right)^\top \\ &= [S(I + P\Theta_{22}^{-1}\Theta_{21})^{-1}P\Theta_{22}^{-1} + \Theta_{12}\Theta_{22}^{-1}]^\top \\ &= [SP\Gamma^{-1} + \Theta_{12}\Theta_{22}^{-1}]^\top \\ &= [(SP + \Theta_{12}\Theta_{22}^{-1}\Gamma)\Gamma^{-1}]^\top. \end{aligned}$$

We now show that the first part reduces to

$$\begin{aligned} SP + \Theta_{12}\Theta_{22}^{-1}\Gamma &= \Theta_{11}P - \Theta_{12}\Theta_{22}^{-1}\Theta_{21}P \\ &\quad + \Theta_{12}\Theta_{22}^{-1}\Theta_{21}P + \Theta_{12} \\ &= \Theta_{11}P + \Theta_{12}. \end{aligned}$$

Consequently,

$$\underline{G}_0 = [(\Theta_{11}P + \Theta_{12})\Gamma^{-1}]^\top = G_0^\top,$$

and hence

$$\mathbf{D}(\Theta) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G_0^\top \\ I \end{bmatrix} \underline{\Gamma} w.$$

By Theorem 2, G_0 and $G_0^\top$ have the same induced L_2 -gain. Hence, for any ρ ,

$$\begin{aligned} \|p_T G_0 \tilde{w}\|_{L_{[a,b]}} &\leq \rho \|p_T \tilde{w}\|_{L_{[a,b]}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0 \\ \iff \|p_T G_0^\top \tilde{w}\|_{L_{[a,b]}} &\leq \rho \|p_T \tilde{w}\|_{L_{[a,b]}} \quad \text{for all } \tilde{w} \text{ and } T \geq 0. \end{aligned}$$

Since $\underline{\Gamma}$ is causally boundedly invertible, the same arguments used for 1) $\implies$ 2) and 2) $\implies$ 1), with G_0 , Γ , w , and $\mathcal{I}$ replaced by $G_0^\top$, $\underline{\Gamma}$, $\underline{w}$, and $\underline{\mathcal{I}}$, respectively, establish 1) $\iff$ 3).

Realization of G_0 . Let $\mathbf{x}$ collect the plant, filter, and any controller states, so that the controller is already included in the primal operators. Partition the filtered realization as

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C}_z & \mathcal{D}_z \\ \mathcal{C}_w & \mathcal{D}_w \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ w \end{bmatrix}.$$

If $\mathcal{D}_w$ is invertible, eliminating w gives the realization

$$G_0 \sim \{\mathcal{T}, \mathcal{A} - \mathcal{B}\mathcal{D}_w^{-1}\mathcal{C}_w, \mathcal{B}\mathcal{D}_w^{-1}, \mathcal{C}_z - \mathcal{D}_z\mathcal{D}_w^{-1}\mathcal{C}_w, \mathcal{D}_z\mathcal{D}_w^{-1}\}, \quad (\text{A.1})$$

with input $\tilde{w}$ and output $\tilde{z}$.

Appendix A.3. Proof of Corollary 1

For $K = 0$ in (4), set $\Lambda := \mathcal{D}_{21}\mathcal{D}_G + \mathcal{D}_{22}$ and define

$$\begin{aligned} (\mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ w \end{bmatrix})(t) &= \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21}\mathcal{C}_G & \mathcal{C}_2 & \Lambda \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix}, \\ (\mathcal{M}_{\Gamma^{-1}} \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix})(t) &:= \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda^{-1}\mathcal{D}_{21}\mathcal{C}_G & -\Lambda^{-1}\mathcal{C}_2 & \Lambda^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ \tilde{w}(t) \end{bmatrix}, \end{aligned}$$

with $\mathcal{M}_\Gamma, \mathcal{M}_{\Gamma^{-1}} \in \Pi_4$ and $\mathcal{M}_{\Gamma^{-1}} = \mathcal{M}_\Gamma^{-1}$. These are coordinate transformations of the states and input. Along system trajectories, they induce $\Gamma := \Theta_{21}P + \Theta_{22} : w \mapsto \tilde{w}$ with $P = 0 \star G$. The dual construction follows from (7). Boundedness of Γ^{-1} is established below. Set $\underline{\Lambda} := \underline{\mathcal{D}}_{21}\mathcal{D}_G^* + \underline{\mathcal{D}}_{22}$ as in Appendix A.2 define $\mathcal{S} = \mathcal{D}_{11} - \mathcal{D}_{12}\mathcal{D}_{22}^{-1}\mathcal{D}_{21}$, its inverse is then given by

$$\underline{\Lambda}^{-1} = [(\mathcal{D}_{11} - \mathcal{D}_{12}\mathcal{D}_{22}^{-1}\mathcal{D}_{21})(I + \mathcal{D}_G\mathcal{D}_{22}^{-1}\mathcal{D}_{21})^{-1}]^* \in \Pi_4. \quad (\text{A.2})$$

Suppose first that the primal LPI holds with $\epsilon > 0$. Define the nonnegative storage $V(t) := \langle \mathcal{T}\mathbf{x}(t), \mathcal{P}\mathcal{T}\mathbf{x}(t) \rangle$. Integrating the LPI from 0 to T , using $\mathbf{x}(0) = 0$, gives

$$V(T) + \mathcal{I} = V(T) + \|p_T G_0 \Gamma w\|_{L_{[a,b]}}^2 - \|p_T \Gamma w\|_{L_{[a,b]}}^2 \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2. \quad (\text{A.3})$$

Since $V(T) \geq 0$, it follows that $\epsilon \|p_T w\|_{L_{[a,b]}}^2 \leq \|p_T \Gamma w\|_{L_{[a,b]}}^2$. For arbitrary $\tilde{w}$, the causal inverse realization supplies $w = \Gamma^{-1}\tilde{w}$ on each finite time interval. Hence

$$\|p_T \Gamma^{-1}\tilde{w}\|_{L_{[a,b]}} \leq \frac{1}{\sqrt{\epsilon}} \|p_T \tilde{w}\|_{L_{[a,b]}} \quad \forall T \geq 0.$$

Thus Γ^{-1} is bounded, with $\|\Gamma^{-1}\| \leq 1/\sqrt{\epsilon}$, and $\mathcal{I} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2$. Applying the same argument on the dual side yields $\underline{\Gamma}^{-1}$ bounded, with $\|\underline{\Gamma}^{-1}\| \leq 1/\sqrt{\epsilon}$ and $\underline{\mathcal{I}} \leq -\epsilon \|p_T \tilde{w}\|_{L_{[a,b]}}^2$. Thus, by Theorem 3, we also find $\mathcal{I} \leq -\epsilon \|p_T w\|_{L_{[a,b]}}^2$ for some $\epsilon > 0$, and by Theorem 1 we find that robust stability and robust performance are guaranteed.

Appendix A.4. Proof of Corollary 2

Write $\mathcal{K} = [\mathcal{K}_G \quad \mathcal{K}_\Theta]$, so that $\mathcal{Z}^* = \mathcal{K}\mathcal{P}$. From (7),

$$\underline{\mathcal{A}} := \underline{\mathcal{A}}_0 + \mathcal{K}^* [\mathcal{B}_u^* \quad 0], \quad \underline{\mathcal{B}} := \underline{\mathcal{B}}_0 + \mathcal{K}^* \mathcal{D}_{zu}^*.$$

Substitution makes (10) the closed-loop dual KYP inequality. Set $\Lambda := \mathcal{D}_{21}\mathcal{D}_G + \mathcal{D}_{22}$. From (4), define

$$\begin{aligned} (\mathcal{M}_\Gamma \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ w \end{bmatrix})(t) &:= \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu}\mathcal{K}_G) & \mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta & \Lambda \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ w(t) \end{bmatrix}, \\ (\mathcal{M}_{\Gamma^{-1}} \begin{bmatrix} \mathbf{x}_G \\ \mathbf{x}_\Theta \\ \tilde{w} \end{bmatrix})(t) &:= \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda^{-1}\mathcal{D}_{21}(\mathcal{C}_G + \mathcal{D}_{zu}\mathcal{K}_G) & -\Lambda^{-1}(\mathcal{C}_2 + \mathcal{D}_{21}\mathcal{D}_{zu}\mathcal{K}_\Theta) & \Lambda^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}_G(t) \\ \mathbf{x}_\Theta(t) \\ \tilde{w}(t) \end{bmatrix}. \end{aligned}$$

For (7), set $\underline{\Lambda} := \underline{\mathcal{D}}_{21}\mathcal{D}_G^* + \underline{\mathcal{D}}_{22}$. (A.2) gives $\underline{\Lambda}^{-1} \in \Pi_4$, so we may define

$$(\mathcal{M}_{\underline{\Gamma}} \begin{bmatrix} \underline{\mathbf{x}}_G \\ \underline{\mathbf{x}}_\Theta \\ \underline{w} \end{bmatrix})(t) := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \underline{\mathcal{D}}_{21}\mathcal{B}_G^* & \underline{\mathcal{C}}_2 & \underline{\Lambda} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_G(t) \\ \underline{\mathbf{x}}_\Theta(t) \\ \underline{w}(t) \end{bmatrix}, \quad (\mathcal{M}_{\underline{\Gamma}^{-1}} \begin{bmatrix} \underline{\mathbf{x}}_G \\ \underline{\mathbf{x}}_\Theta \\ \underline{\tilde{w}} \end{bmatrix})(t) := \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\underline{\Lambda}^{-1}\underline{\mathcal{D}}_{21}\mathcal{B}_G^* & -\underline{\Lambda}^{-1}\underline{\mathcal{C}}_2 & \underline{\Lambda}^{-1} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_G(t) \\ \underline{\mathbf{x}}_\Theta(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}.$$

These PI maps induce Γ and $\underline{\Gamma}$ in Theorem 3 with $P = K \star G$. By following the same argument as in the proof of Corollary 1 in Appendix A.3, we find that robust stability and robust performance are guaranteed.

Appendix A.5. Proof of Lemma 1

Suppose that the dual IQC

$$\left\langle p_T \begin{bmatrix} P_\Delta^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{Y}_\Delta} \begin{bmatrix} P_\Delta^\top \underline{w}_\Delta \\ \underline{w}_\Delta \end{bmatrix} \right\rangle_{L_{[a,b]}}$$

is parametrized according to (14). We need to show that Δ satisfies the primal IQC. First, observe the dual IQC can be factorized as

$$\mathcal{Y}_\Delta = \frac{1}{2} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \underbrace{\begin{bmatrix} \mathcal{S} & I \\ \mathcal{S} & -I \end{bmatrix}}_{\underline{\Theta}_1} \underbrace{\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}}_{\underline{\Theta}_2} \quad (\text{A.4})$$

Them, we use the multiplicative and involutive property of the dual operation such that $\mathbf{D}(\Theta_1\Theta_2) = \mathbf{D}(\Theta_1)\mathbf{D}(\Theta_2)$ for any two invertible Θ_1 and Θ_2 and $\mathbf{D}(\mathbf{D}(\Theta)) = \Theta$ to recover the primal IQC. Since $\underline{S}_0 > 0$, $\underline{S}$ is closed under inversion with $\underline{S}_0^{-1} > 0$. Applying these properties to the two factors of $\underline{\Theta}$ in (A.4) gives

$$\begin{aligned}\Theta &= \mathbf{D}(\underline{\Theta}) = \mathbf{D}\left(\frac{1}{\sqrt{2}} \begin{bmatrix} I & \underline{S} \\ I & -\underline{S} \end{bmatrix}\right) \mathbf{D}\left(\begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}\right) \\ &= \frac{1}{\sqrt{2}(\beta - \alpha)} \begin{bmatrix} -\underline{S}^{-1} & -I \\ -\underline{S}^{-1} & I \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}.\end{aligned}$$

We omit $\frac{1}{(\beta - \alpha)} > 0$ therefore,

$$\mathcal{V}_\Delta = \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}^* \begin{bmatrix} 0 & \underline{S}^{-1} \\ \underline{S}^{-1} & 0 \end{bmatrix} \begin{bmatrix} \beta & -1 \\ -\alpha & 1 \end{bmatrix}$$

For completion we use the proof in [32] Then, for all $s \in [a, b]$ and $t > 0$, we have that

$$(\beta z_\Delta(t, s) - \phi(z_\Delta(t, s)))(\phi(z_\Delta(t, s)) - \alpha z_\Delta(t, s)) \geq 0,$$

and hence

$$(\beta z_\Delta(t, s) - \phi(z_\Delta(t, s)))\underline{S}_0^{-1}(s)(\phi(z_\Delta(t, s)) - \alpha z_\Delta(t, s)) \geq 0.$$

Therefore, Δ satisfies the IQC defined by $\underline{\Psi}_\Delta = I$ and (14).